



AutoCAD
ELECTRICAL

COURSE SLIDES · WSQ

Technical Drawing with AutoCAD Electrical

WSQ Course Code: TGS-2023037466

TSC: Technical Drawing · DSN-TDR-4005-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 6.1 · 21 August 2026

Trainer: _____ · www.tertiarycourses.com.sg · enquiry@tertiaryinfotech.com · +65 6100 0613

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS

PhD — data science, engineering design and CAD/CAE practice.

DELIVERS

WSQ courses on technical drawing (AutoCAD, AutoCAD Electrical, Revit, Fusion 360) and engineering software.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with AutoCAD or electrical design (if any).
- What you want to be able to draw or document after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is stupid.

3

Respect each other's views.

4

Exit silently for calls or toilet breaks.

5

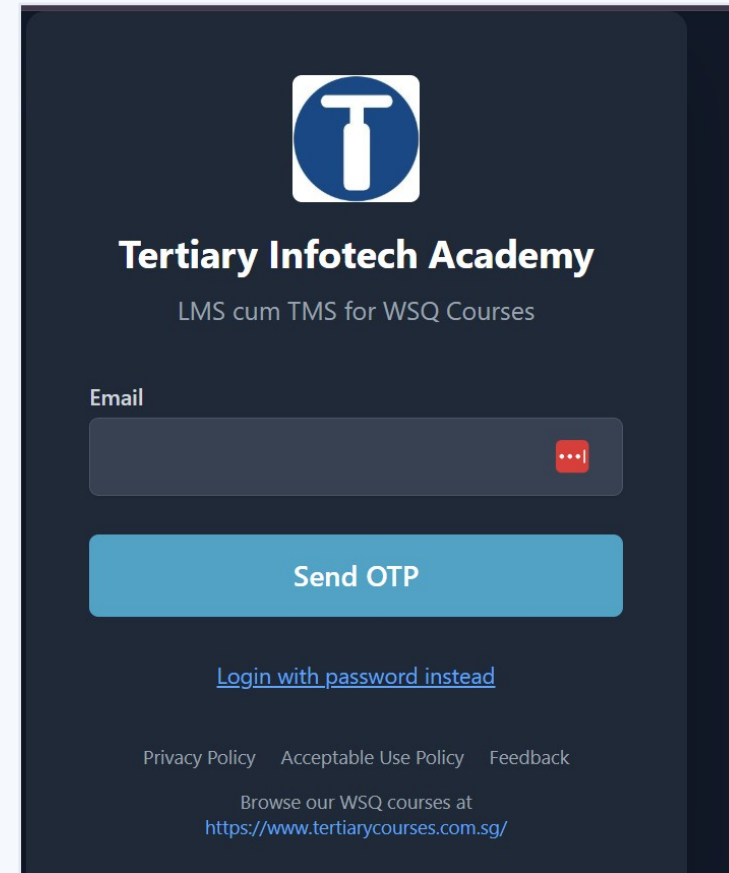
Be punctual; return from breaks on time.

6

75% attendance is required for funding.

Download Course Material

- All course materials live on the Tertiary Infotech LMS/TMS portal.
- Portal: <https://lms-tms.tertiaryinfotech.com>
- Log in with the credentials sent to your registered email.
- Download the slides and Learner Guide — you may use them during the open-book assessment.
- Your attendance records, assessment submission and certificate are also on the portal.



The screenshot shows the login interface for the Tertiary Infotech Academy. At the top is a logo consisting of a white 'T' inside a blue circle. Below the logo, the text 'Tertiary Infotech Academy' is displayed in a bold, white font, followed by 'LMS cum TMS for WSQ Courses' in a smaller, lighter font. The main section features an 'Email' label above a dark grey input field. To the right of the input field is a red button with three white dots. Below the input field is a large, light blue button labeled 'Send OTP'. Underneath this button is a link that says 'Login with password instead'. At the bottom of the page, there are three links: 'Privacy Policy', 'Acceptable Use Policy', and 'Feedback'. Finally, there is a line of text that says 'Browse our WSQ courses at' followed by the URL 'https://www.tertiarycourses.com.sg/'.

Lesson Plan — 2 Days

Day 1

- Day 1 — Getting started, projects and schematic drawing
 - Digital attendance (AM) · introductions · learning outcomes · course outline
 - Topic 1: Introduction to Technical Drawing Using AutoCAD Electrical (Labs 1–2)
 - Lunch break · digital attendance (PM)
 - Topic 2: Analysing Technical Drawings (Labs 3–4)
 - End of Day 1

Day 2

- Day 2 — Analysis, documentation, submission and assessment
 - Digital attendance (AM) · Day 1 recap
 - Topic 2 (continued): cross-referencing & review (Lab 5)
 - Topic 3: Reviewing Drawings for Strategic Relevance & Standard Compliance (Labs 6–8)
 - Course feedback & TRAQOM survey · digital attendance (Assessment)
 - Final Assessment: WA (SAQ) 1 hr + PP 1 hr

Skills Framework

- TSC Title: Technical Drawing
- TSC Code: DSN-TDR-4005-1.1
- This course is mapped to the Skills Framework for Design — Technical Drawing TSC.
- The assessment covers every Knowledge (K1–K6) and Ability (A1–A6) statement of the TSC.

TSC Knowledge Statements

1

K1

Symbols, standards and conventions used in technical drawings across various fields and regions

2

K2

Methods of technical drawing by instruments and computer-aided design (CAD) systems

3

K3

Types of technical drawings

4

K4

Technical drawing standards

5

K5

Fundamentals of drafting design documentation

6

K6

Concepts of information technology (IT) to facilitate the usage of technical drawing software

TSC Ability Statements

1

A1

Devise metrics and guidelines for the production of technical drawings

2

A2

Formulate solutions for problems encountered during the development of technical drawings

3

A3

Review technical drawings for accuracy and adherence to established specifications

4

A4

Analyse drawings to determine their construction needs to deliver upon design solutions

5

A5

Review technical drawings and/or changes to them, to ensure relevance to the organisation's strategies

6

A6

Review drawings and specifications to ensure conformance to requirements of relevant standards for submission

Learning Outcomes

1

LO1 — Devise & solve

Devise metrics for technical drawing production with AutoCAD Electrical and formulate solutions for development problems. (K1, K2, A1, A2)

2

LO2 — Analyse & review

Analyse technical drawings to determine construction needs while reviewing accuracy and adherence to specifications. (K3, K4, A3, A4)

3

LO3 — Strategy & standards

Review drawings for relevance to the organisation's strategies and conformance to submission standards. (K5, K6, A5, A6)

Course Outline

Topic 1 · Introduction

- AutoCAD Electrical user interface
- Symbols, standards & conventions
- Ribbon, palettes & navigation
- Projects & drawing management
- Labs 1–2

Topic 2 · Analysing Drawings

- Types of schematic drawings
- Components, terminals & PLC symbols
- Wires, numbers & signal arrows
- Cross-referencing & review tools
- Labs 3–5

Topic 3 · Review & Compliance

- Drafting design documentation
- Panel layouts & part-list reports
- Custom symbols
- Plot & publish for submission
- Labs 6–8

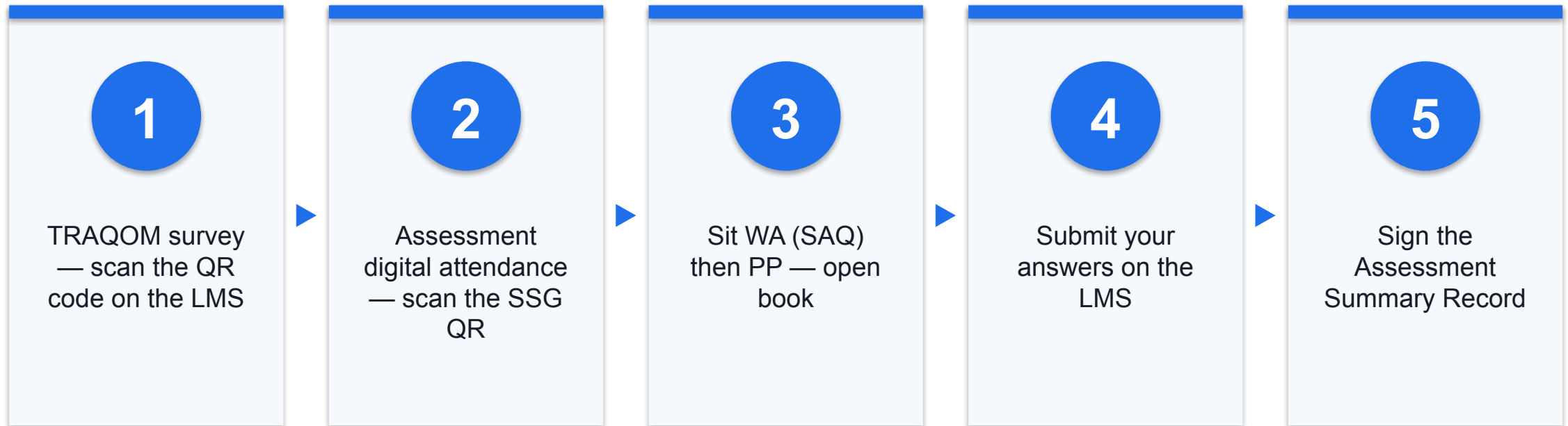
Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. (K1–K6)
- Practical Performance (PP) — hands-on AutoCAD Electrical tasks, 1 hour, open book. (A1–A6)
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

Assessment Flow



Criteria for Funding

- Minimum attendance rate of 75% based on the SSG Digital Attendance record.
- Complete the assessment and be assessed as 'Competent'.

TOPIC 01

Introduction to Technical Drawing Using AutoCAD Electrical

User interface · symbols, standards & conventions · ribbon · navigation · projects & drawings (K1, K2 · A1, A2)

Key Concepts — Topic 1

1

Symbols, standards, conventions

IEEE 315/315A, IEC-60617 and NFPA symbol libraries make an electrical drawing readable across fields and regions. (K1)

2

CAD vs manual drawing

AutoCAD Electrical automates tagging, wire numbering and cross-referencing that instrument drawing does by hand. (K2)

3

The AutoCAD Electrical UI

Ribbon tabs, Quick Access Toolbar, marking menu, palettes, status bar, command line and navigation tools.

4

Projects drive everything

A .wdp project file groups interrelated drawings so project-wide functions can retag, renumber and report.

5

Devise metrics & guidelines

Templates, drawing properties, reference systems and title blocks standardise production. (A1)

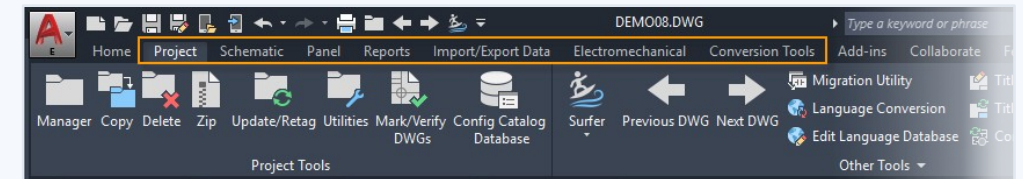
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Formulate solutions

Grid/snap, ortho, object snap and the Circuit Clipboard solve day-to-day drawing development problems. (A2)

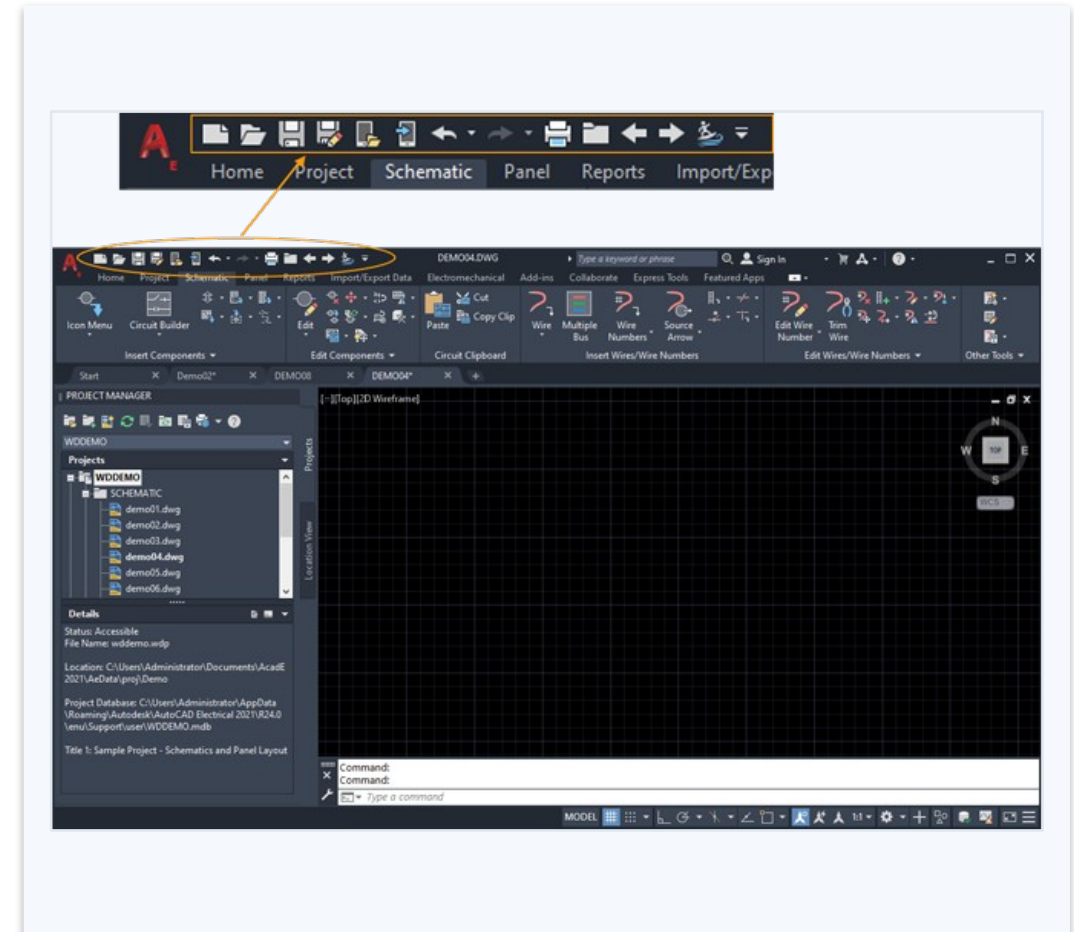
Ribbon Tabs and Panels

- AutoCAD Electrical presents a standard tabbed ribbon across the top of the drawing area.
- Electrical-specific tabs (Schematic, Panel, Reports) sit alongside the core AutoCAD tabs.
- Each tab is divided into panels that group related commands.
- Right-click in the tabs area to choose which tabs and panels to display or hide.



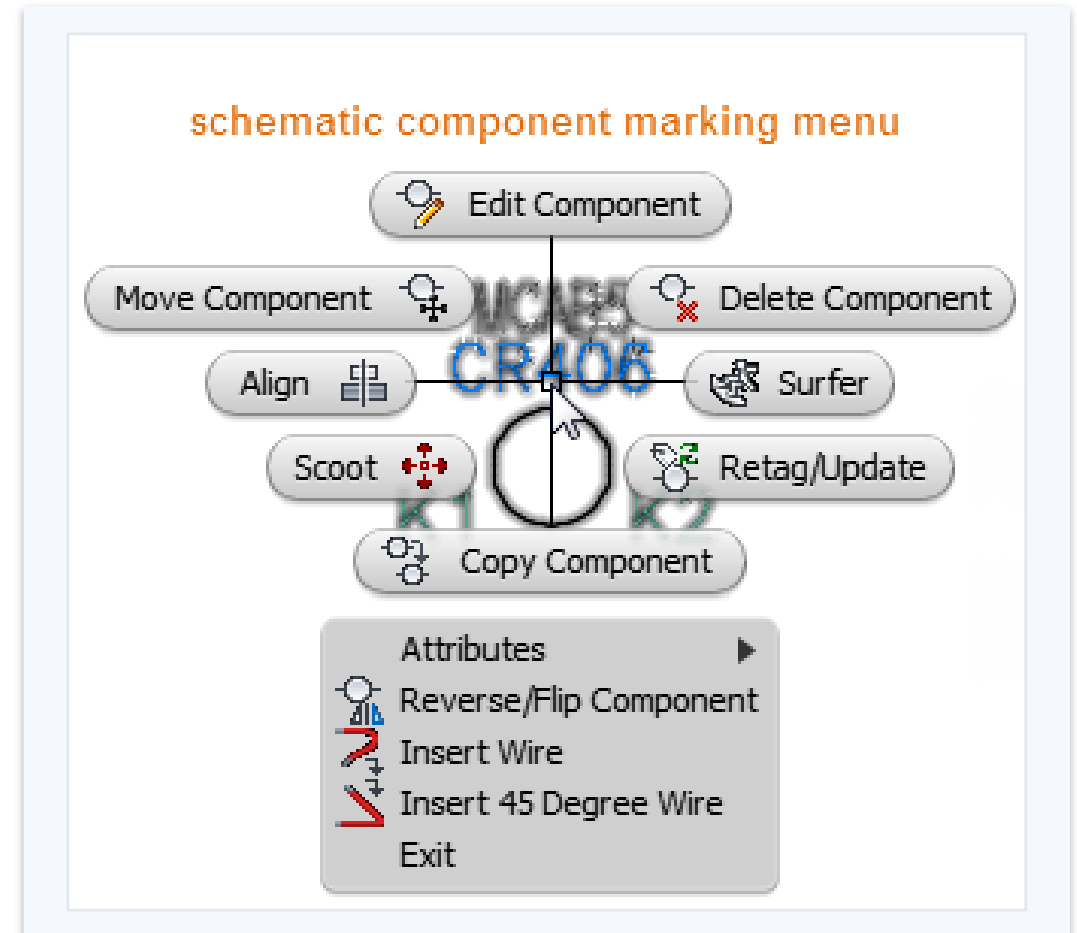
Quick Access Toolbar

- The Quick Access Toolbar keeps New, Open, Save, Plot and Undo one click away.
- It also carries navigation commands used frequently in AutoCAD Electrical.
- Customise it: click the drop-down at its right end and tick the commands you use most.



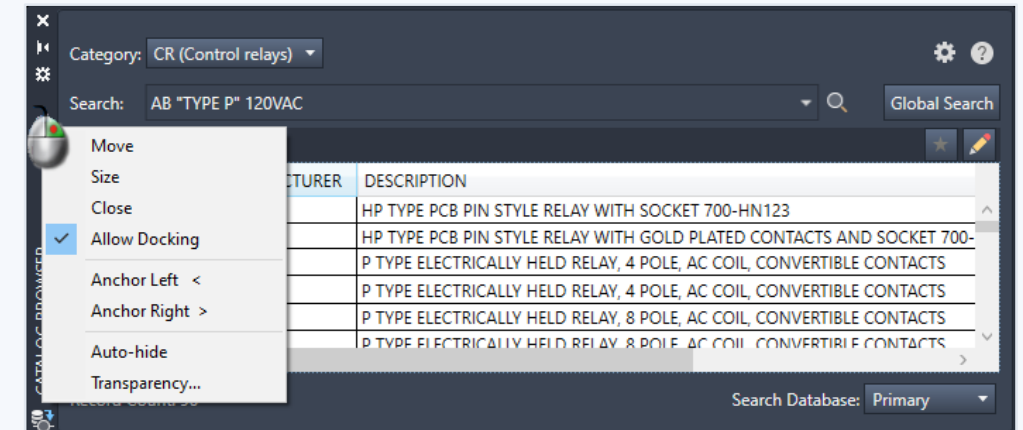
Marking Menu

- The marking menu is a radial context menu: hover an electrical object and right-click.
- The options shown depend on the type of object under the cursor — click one to run it.
- Power use: press-and-hold right-click, move the cursor toward the desired option, release — a gesture runs the command directly.



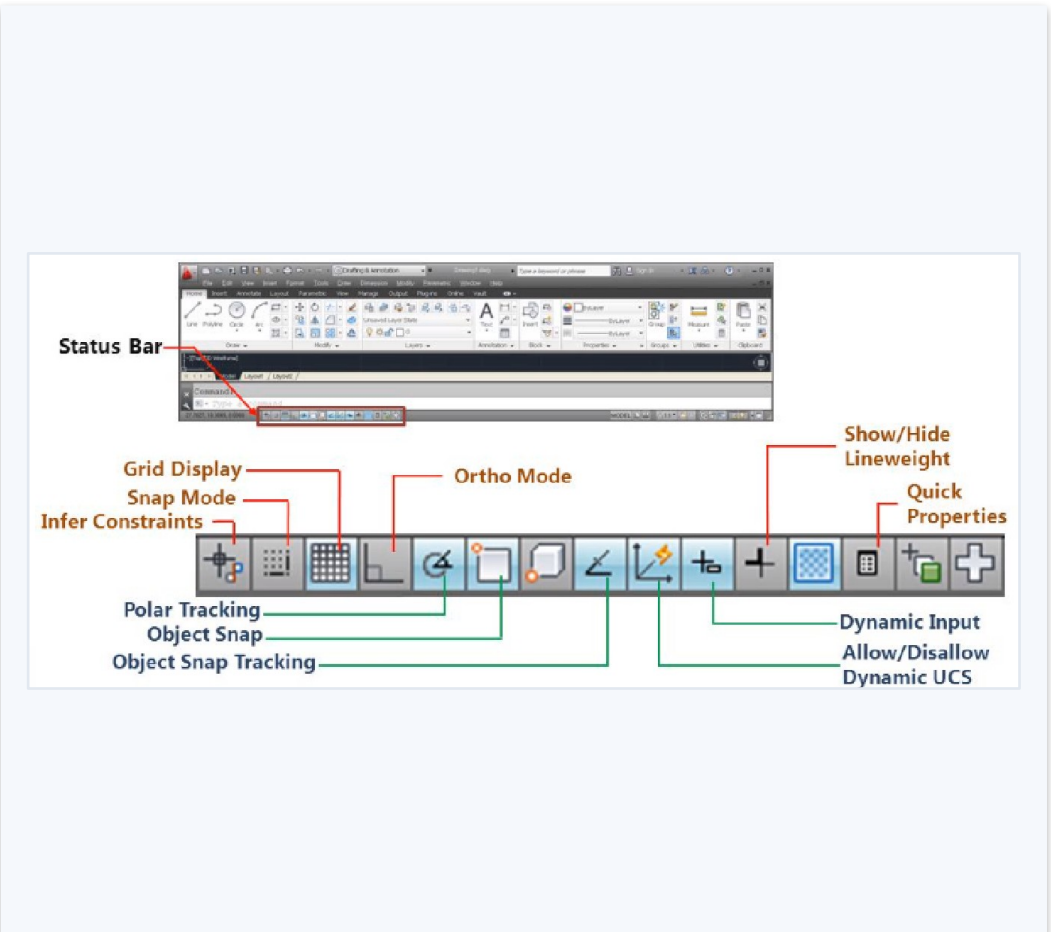
Palettes

- Palettes manage projects (Project Manager) and the catalog database (Catalog Browser).
- A palette stays on screen while other commands run — always accessible.
- Dock, resize or set auto-hide to control how much drawing space a palette uses.



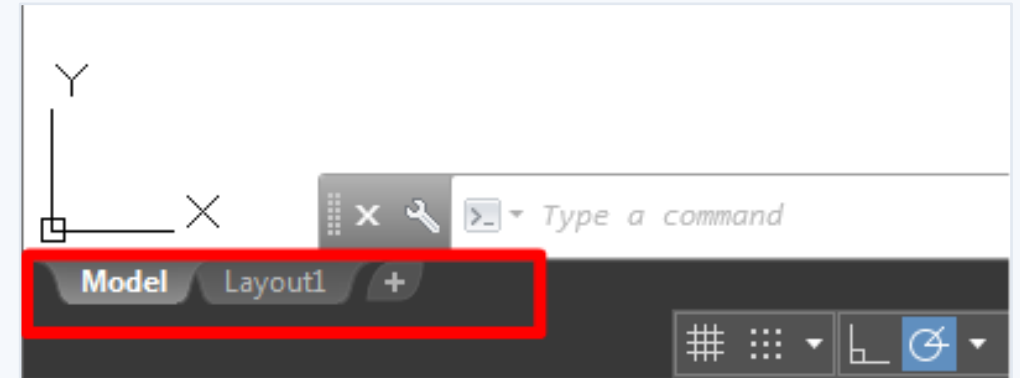
Status Bar — Precision Settings

- The status bar toggles the settings that make a drawing accurate and precise.
- Grid (F7) displays the reference grid; Snap (F9) locks the cursor to grid points.
- Ortho (F8) constrains movement to horizontal/vertical; Polar tracks set angles.
- 2D Object Snap (F3) jumps to endpoints, midpoints and circle centres as you draw.
- These settings are your first production metric: set them before you draw.



Model Space and Paper Space

- Model space is a limitless drawing area where you draw at 1:1 real-world size.
- First decide what one unit represents (1 mm, 1 cm, 1 inch ...) — then draw to that scale.
- Paper space (layouts) arranges scaled views of the model on a titled sheet for plotting.

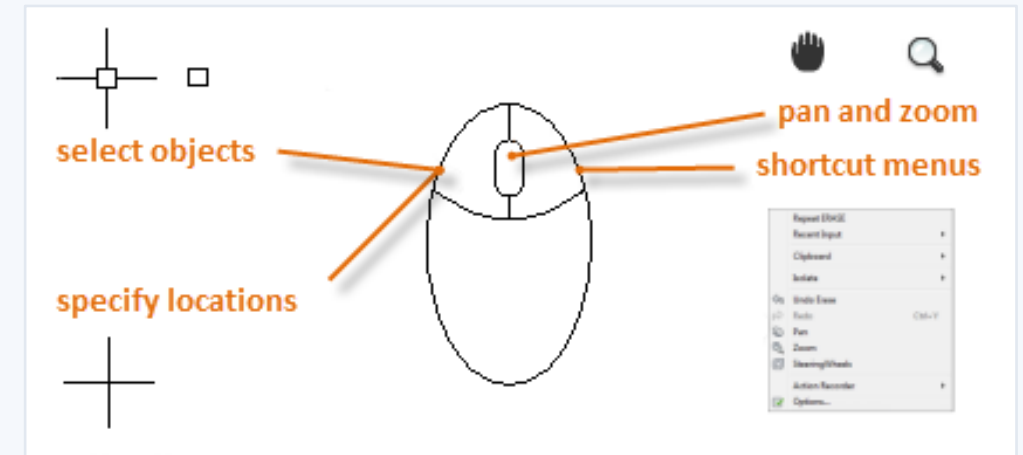


Command Line

- The Command window accepts commands and system variables and prompts you through each command sequence.
- Every ribbon click has a typed equivalent — experienced drafters mix both.
- Ctrl+9 toggles the command window display.
- Watch the prompts: they tell you exactly what input the command expects next.

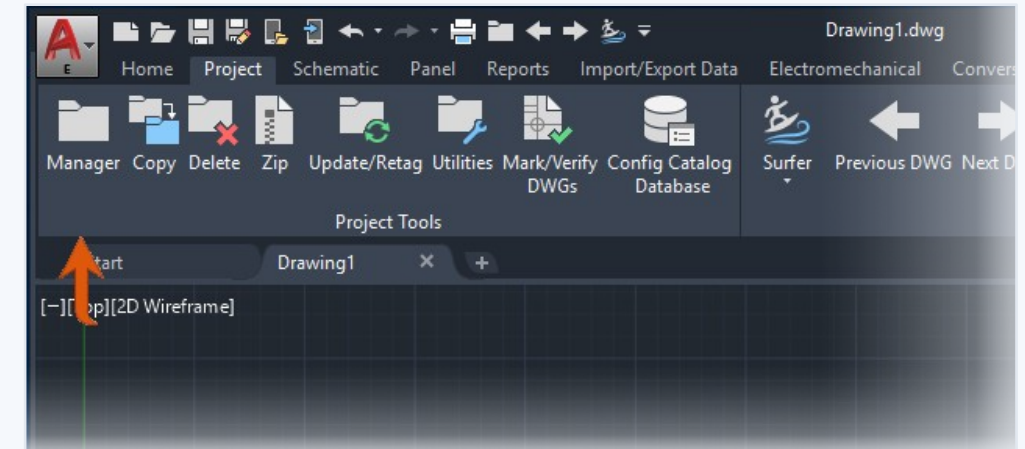
Navigation Tools

- Pan moves the view parallel to the screen; Zoom changes magnification; Orbit rotates 3D views.
- Mouse shortcuts: press the scroll wheel to Pan, roll it to Zoom, double-click it to Zoom Extents.
- ZOOM (Z) options: All shows everything, Window zooms a picked area, Previous restores up to 10 earlier views, Scale magnifies by factor.



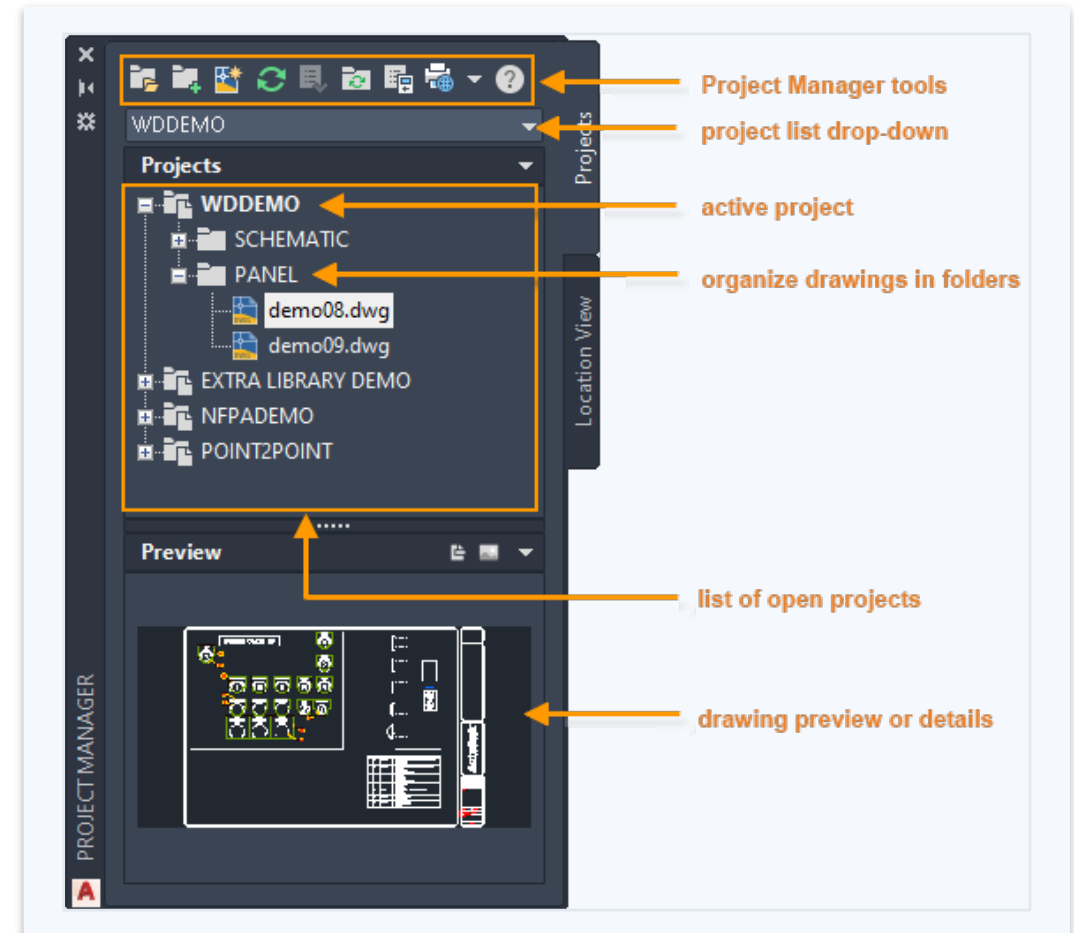
Projects — the .wdp File

- AutoCAD Electrical is project-based: a project is a set of interrelated drawings.
- All project information — including the drawing list — is saved in a .wdp project file.
- Project Manager creates, opens and manages projects and their settings.
- Working project-wide is what lets tags, wire numbers and cross-references stay consistent.



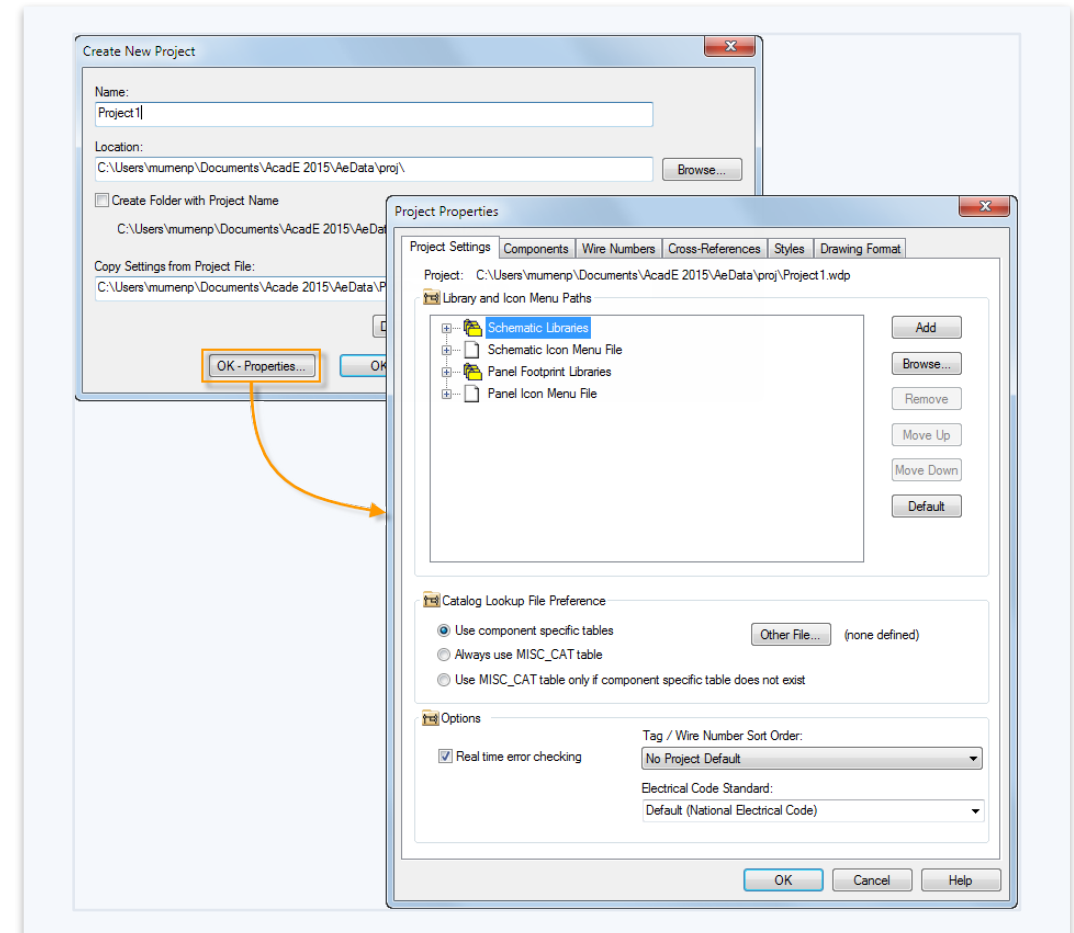
Project Manager

- Project Manager is a palette — it can stay on screen while you use other commands.
- It can be docked, resized and set to auto-hide.
- Create a project and the .wdp file lands in your chosen folder; the new project becomes active and you add drawings to it.



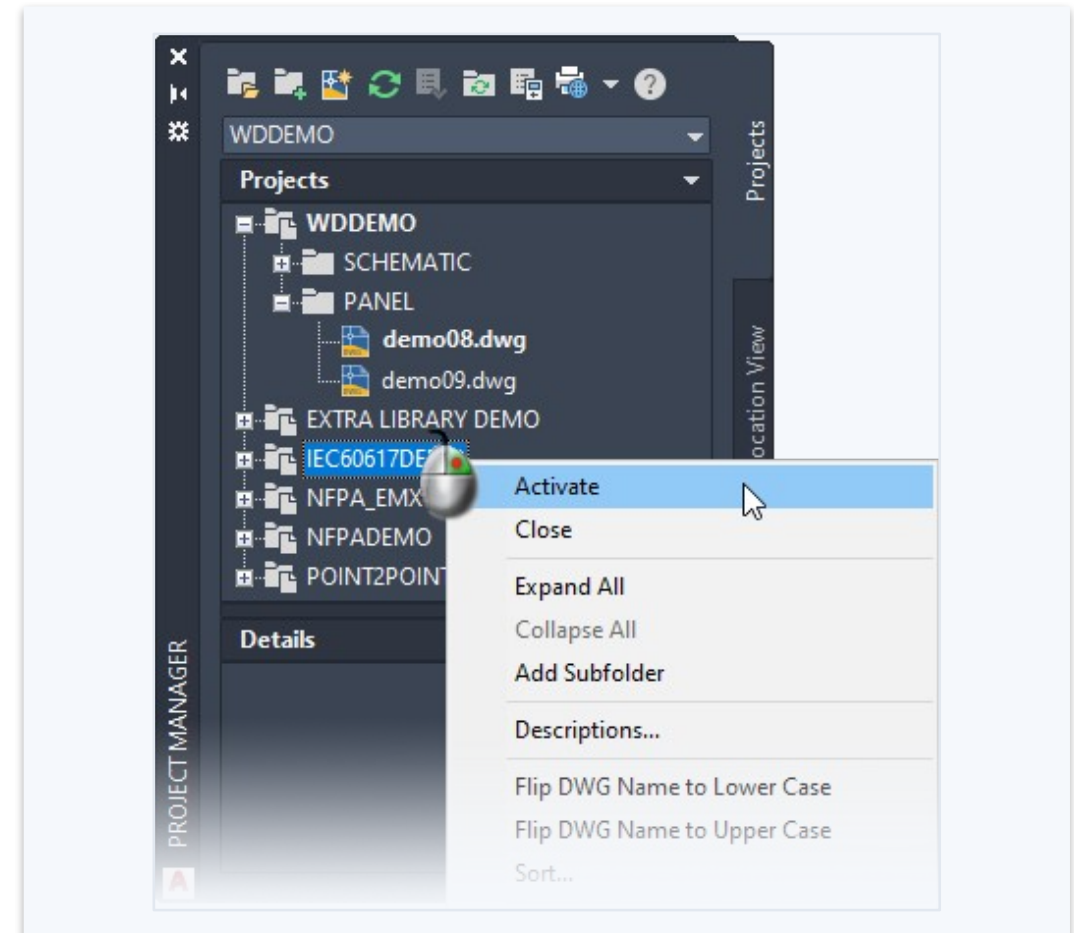
Project Properties

- Define library paths, drawing format and reference style when you start a project — or any time later.
- Two kinds of properties: drawing defaults (copied into each new drawing) and project-wide settings.
- Setting properties once at project level is how an organisation standardises its drawings.



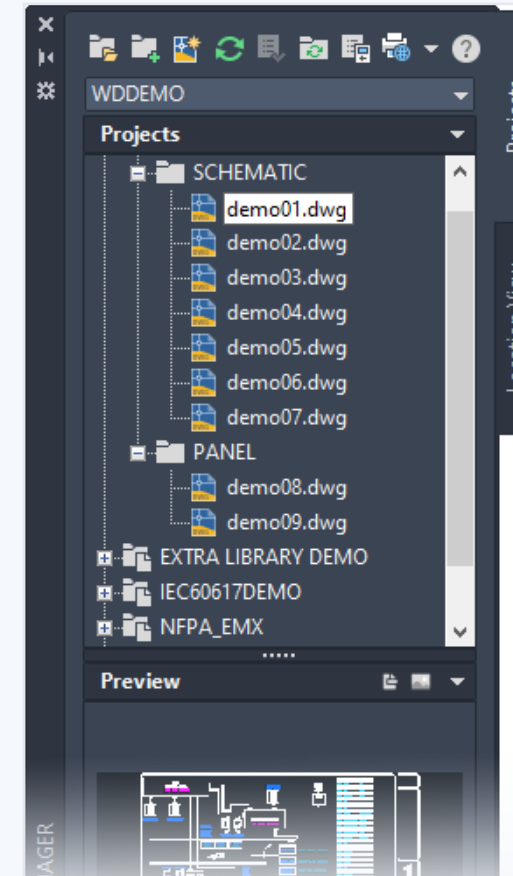
Activate a Project

- Activate from the list of open projects in Project Manager.
- Or browse to the project's .wdp file, or pick from the recent-projects list.
- An opened project joins the Project Manager list and becomes the active project.
- Copy an existing project to reuse its properties, drawings and support files in a new job.



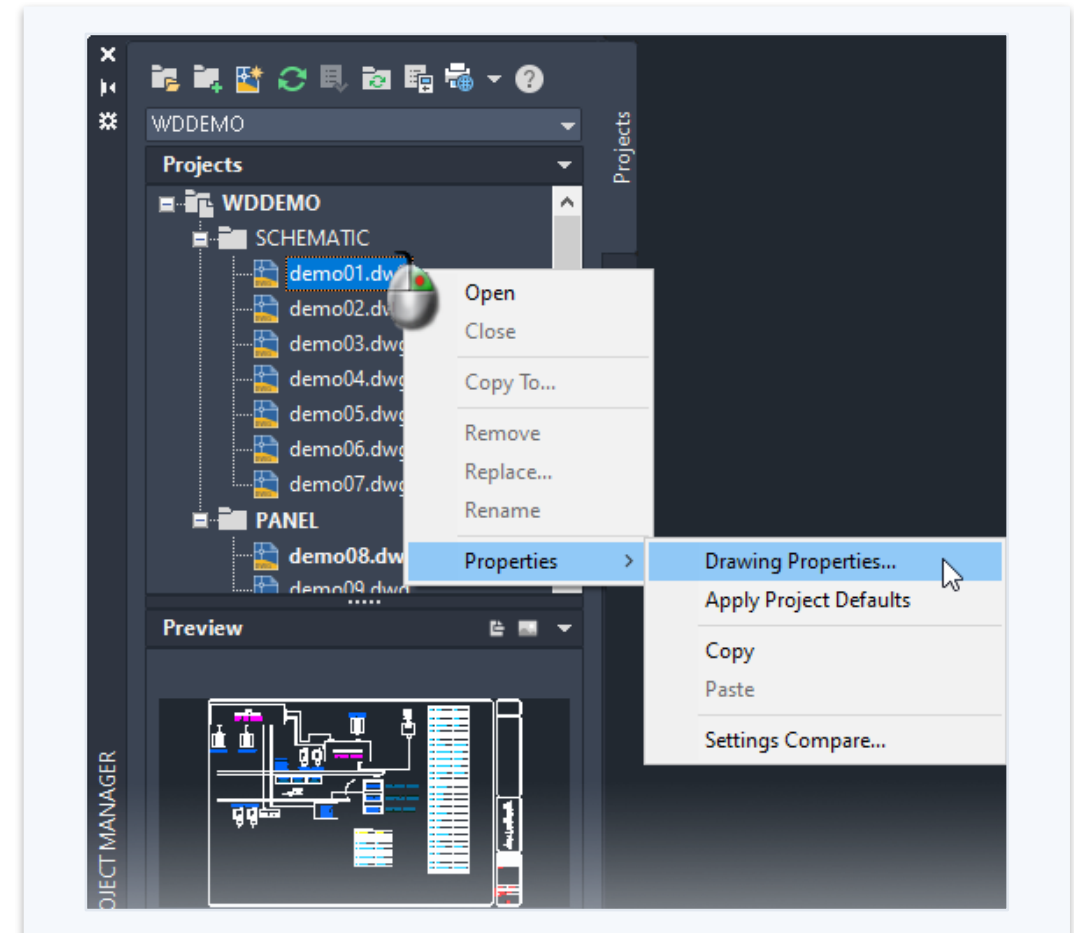
Drawings in a Project

- The drawing order sets the processing order for project-wide functions — drag and drop to reorder.
- Add folders within a project to organise drawings.
- New Drawing creates a drawing inside the active project; give it a template with predefined electrical properties, wire layers, a reference system and a title block.



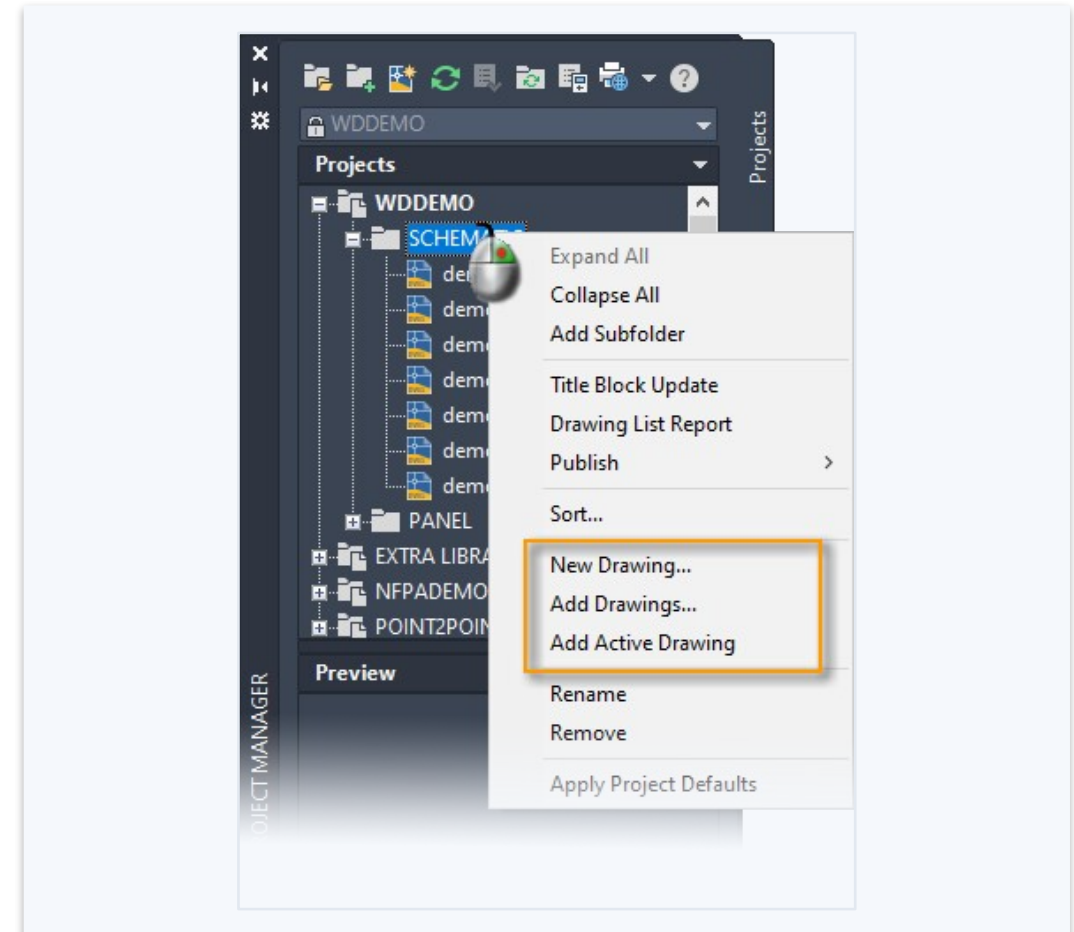
Drawing Properties

- Drawing Properties control tagging and referencing per sheet: sheet values, installation and location codes.
- Reference system: ladder, X-Y Grid or X Zones; formats for component tags, wire numbers and cross-references.
- Styles: PLC modules, wire crossing and wire connection appearance.
- Properties come from the template or the project defaults — override via OK-Properties when creating the drawing.



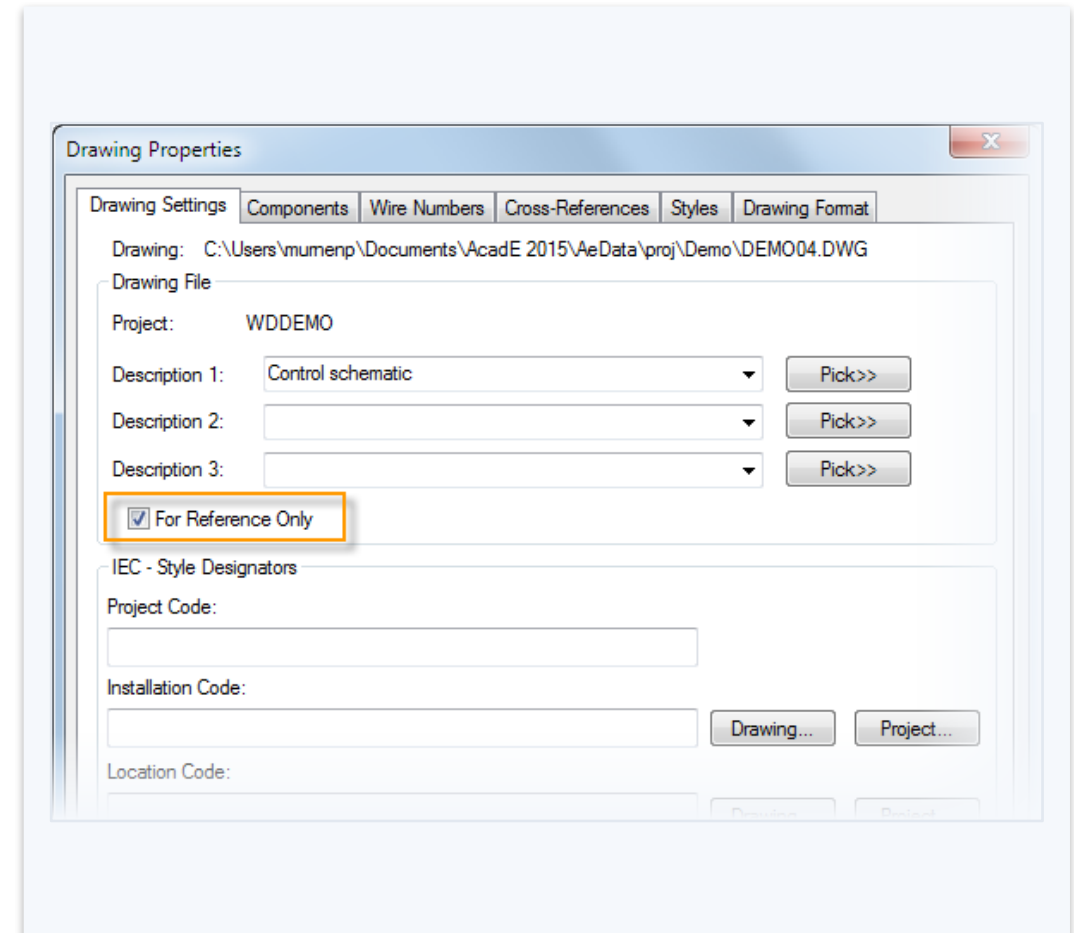
Copy and Add Drawings

- Copy a single drawing from any open project into the active project via Project Manager.
- Add existing drawings any time — the way to reuse sheets from a previous job.
- After adding, run Project-Wide Update/Retag to refresh ladder references, component tags, wire numbers and cross-references.



Reference-Only Drawings

- Some drawings belong to the project but must not be processed by project-wide functions.
- Mark them Reference Only in the drawing properties — tagging, cross-referencing and reporting skip them.
- Typical examples: cover sheets, general notes, mechanical outlines.

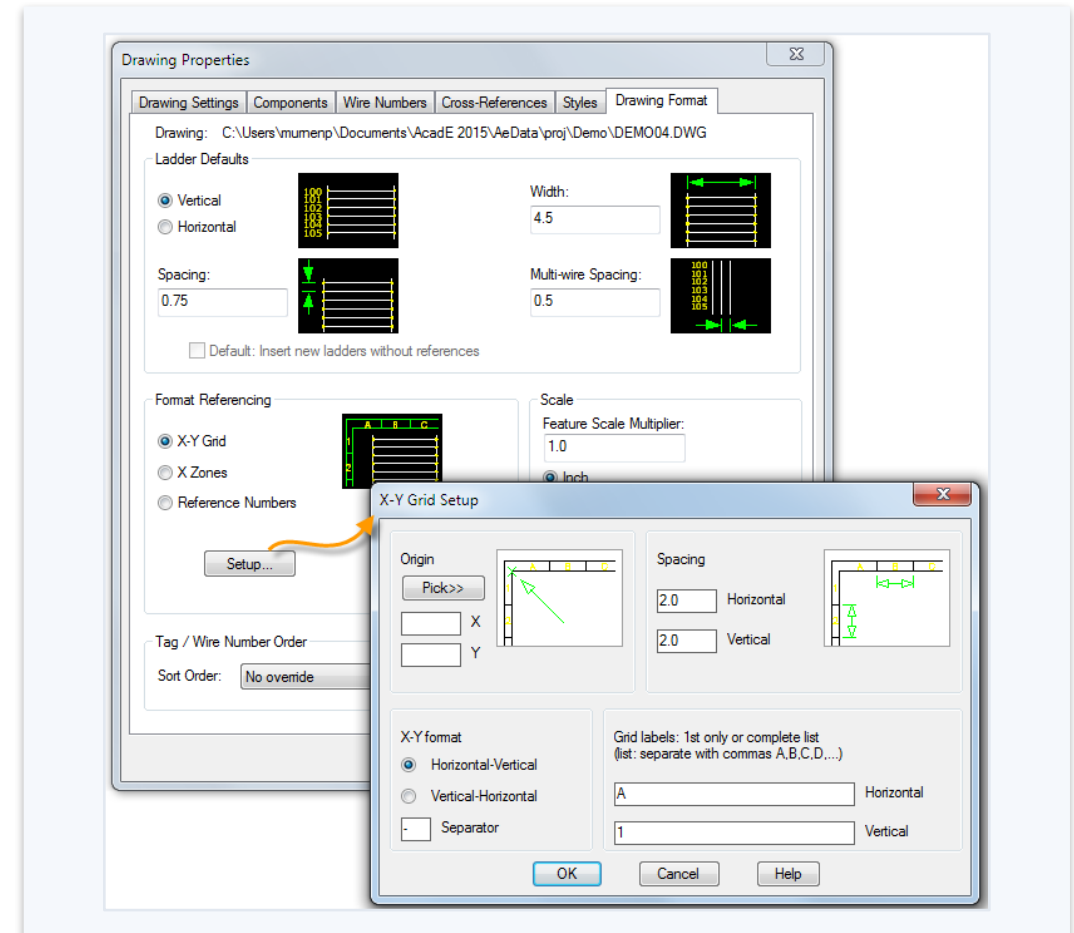


Title Block Setup

- Insert a .dwg as the title block and link project/drawing property values to its attributes.
- Title Block Setup creates and defines the mapping between properties and attributes.
- Once mapped, sheet number, description and revision fill in automatically on every sheet — a core drafting standard.

Reference Systems — Ladders, X-Y Grid, X Zones

- Each schematic drawing can define a reference system: ladder rungs, X-Y grid or X zones.
- The reference system assigns unique component tags and wire numbers (unless sequential tagging is used).
- It also drives cross-referencing between parents and children.
- Insert the ladder or grid from Schematic → Insert Wires/Wire Numbers; define it on a template to save time.



Hands-On Labs — Topic 1

Lab 1

- Set Up a New AutoCAD Electrical Drawing
- A correctly configured metric drawing sheet, saved as a reusable baseline for the rest of the course.

Lab 2

- Create a Project with Project Manager
- A verified working copy of an AutoCAD Electrical .wdp project, with its schematic and panel sheets in the correct order and current library paths resolved.

Set Up a New AutoCAD Electrical Drawing

LAB 1

Create a new drawing from the ACADISO template, then set the workspace, drawing limits and units — the production metrics every drawing in your organisation should start from.

You'll build

A correctly configured metric drawing sheet, saved as a reusable baseline for the rest of the course.

Tools: AutoCAD Electrical · ACADISO template · Status bar · Grid & Snap

Set Up a New AutoCAD Electrical Drawing

Test it

ZOOM All shows the full 420×297 sheet, the crosshair snaps in 10 mm increments, and STATUS reports decimal units — your baseline metrics are in force.

Create a Project with Project Manager

LAB 2

AutoCAD Electrical is project-based: a .wdp project groups interrelated schematic and panel drawings so project-wide functions can retag, renumber, cross-reference and report. Inspect the supplied AutoCAD Electrical sample project, then create a safe working copy for the connected labs.

You'll build

A verified working copy of an AutoCAD Electrical .wdp project, with its schematic and panel sheets in the correct order and current library paths resolved.

Tools: Project Manager · .wdp/.aepx project files · linked schematic and panel DWGs

Create a Project with Project Manager

Test it

Project Manager shows **TRAINING** in bold, every sample-project drawing resolves from the working folder, no unresolved legacy library path remains, and the original supplied package is unchanged.

Recap — Topic 1

- A1 — devise metrics and guidelines for technical drawing production (template, workspace, limits, units) (Lab 1)
- A1, A2 — standardise production with a project; solve multi-drawing organisation problems (Lab 2)
- Key concepts: Symbols, standards, conventions, CAD vs manual drawing, The AutoCAD Electrical UI, Projects drive everything, Devise metrics & guidelines, Formulate solutions.

Lunch Break

Back at 1300 · PM digital attendance

TOPIC 02

Analysing Technical Drawings

Types of schematic drawings · components, terminals & PLC symbols · wires & part lists · accuracy review (K3, K4 · A3, A4)

Key Concepts — Topic 2

1

Types of technical drawings

Schematics, panel layouts, wiring diagrams, one-line diagrams and reference-only sheets each serve a construction purpose. (K3)

2

Technical drawing standards

Ladder references, X-Y grids, wire-number formats and cross-reference styles are the standards drawings are checked against. (K4)

3

Schematic components

Insert from the icon menu or catalog database; annotate with tags, descriptions and catalog data.

4

Terminals & PLC symbols

Standard part lists rely on terminal strips and parametric PLC I/O modules that adapt to the ladder.

5

Wires, numbers & arrows

Wire layers, wire numbers and source/destination signal arrows carry the circuit across sheets.

6

Review for accuracy

Cross-referencing, Surfer and Mark/Verify audit a drawing set for accuracy and adherence to specifications. (A3, A4)

Types of Technical Drawings

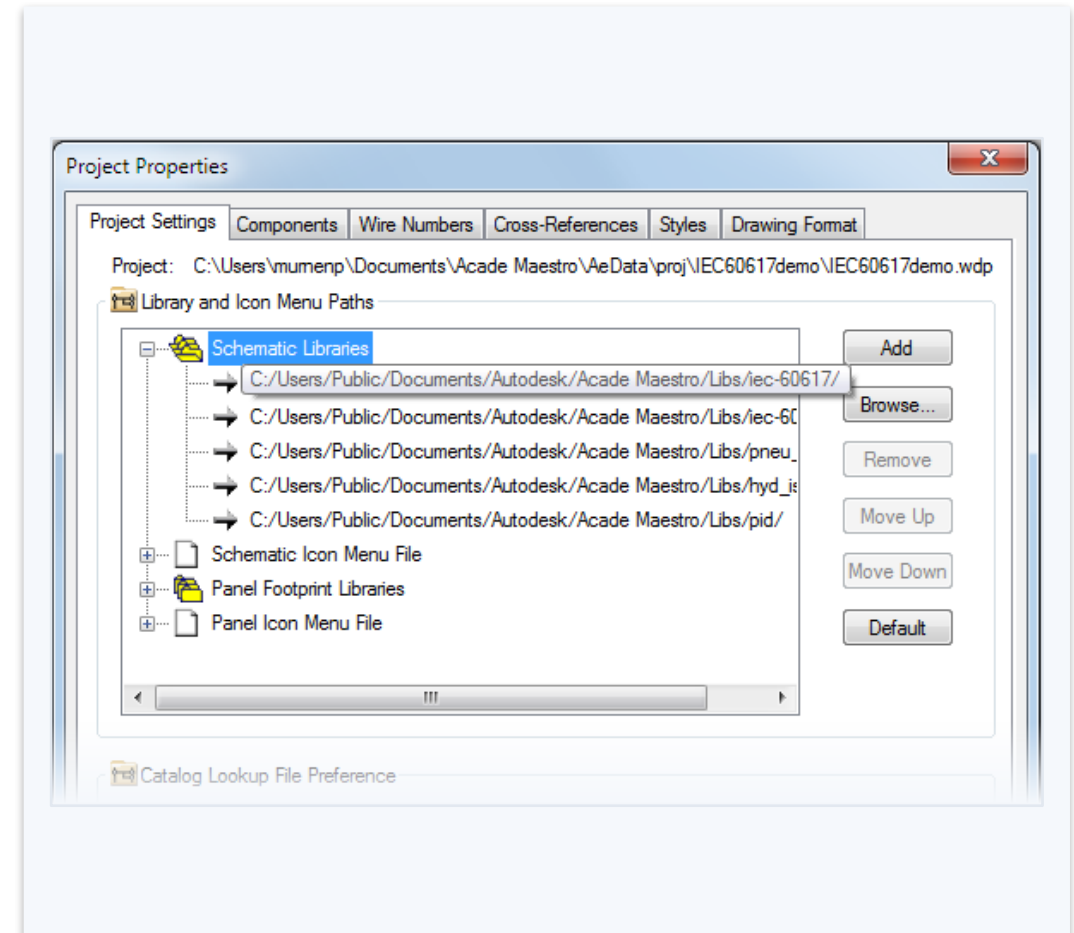
- Schematic diagrams show circuit logic: ladders, components, wires and signals.
- Panel layout drawings show the physical arrangement of components in an enclosure.
- Wiring / connection diagrams document point-to-point conductor runs and terminal strips.
- One-line (single-line) diagrams summarise power distribution at system level.
- Reference-only sheets (covers, notes, outlines) support the set without carrying circuit data.
- Each type answers a different construction need — a full drawing set uses them together.

Schematic Components — Grid and Snap Discipline

- Insert schematic components, generate PLC modules, and insert or copy circuits.
- Work with grid lines on and snap-to-grid when inserting components.
- GRID sets grid spacing; SNAP sets snap spacing; F7 toggles grid visibility, F9 toggles snap.
- Consistent snap spacing keeps rungs aligned so wires connect cleanly — an accuracy standard.

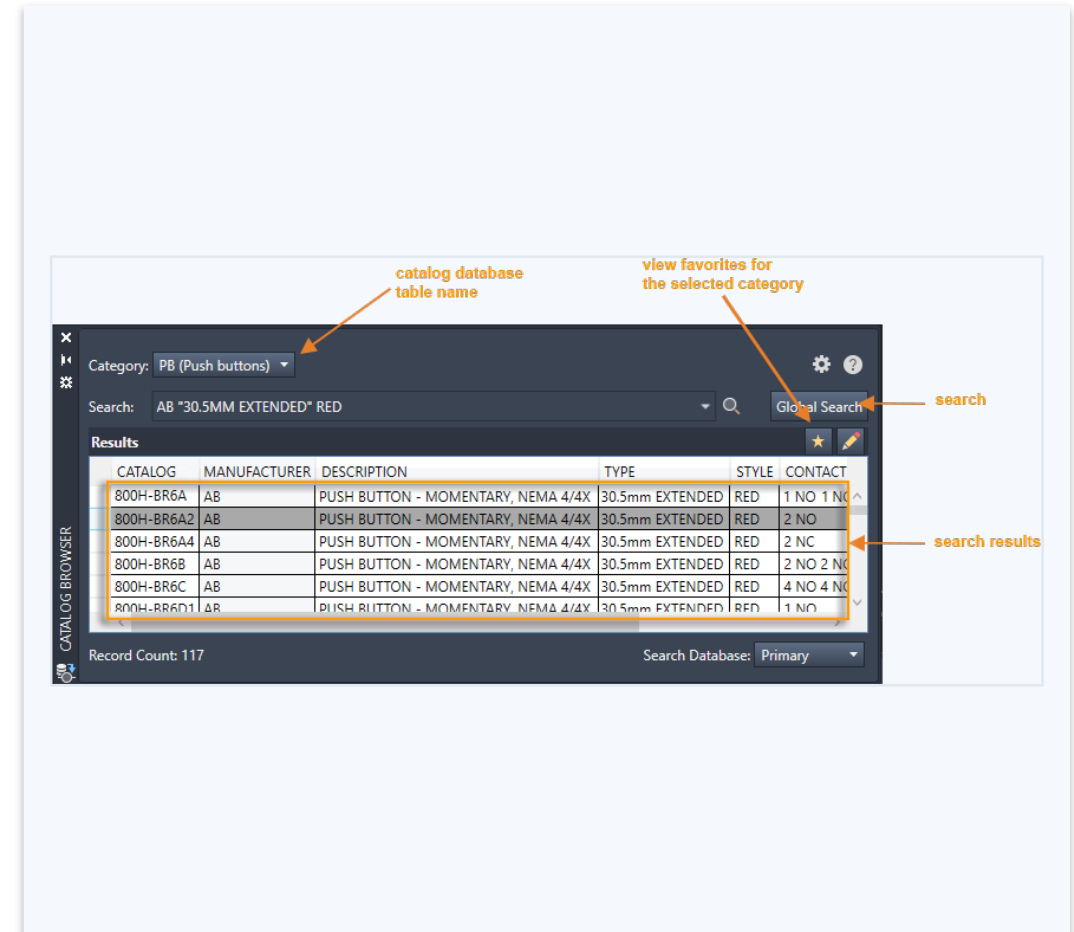
Symbol Library Standards

- AutoCAD Electrical ships symbol libraries complying with IEEE 315/315A, IEC-60617 and NFPA.
- IEEE and IEC libraries are metric; NFPA library symbols are in inches (scale with the Modify-library utility).
- The old JIC standard was absorbed into NFPA 79; JIC and older IEC libraries remain for legacy support.
- Pick ONE standard per project in Project Properties — mixing standards breaks readability across regions.



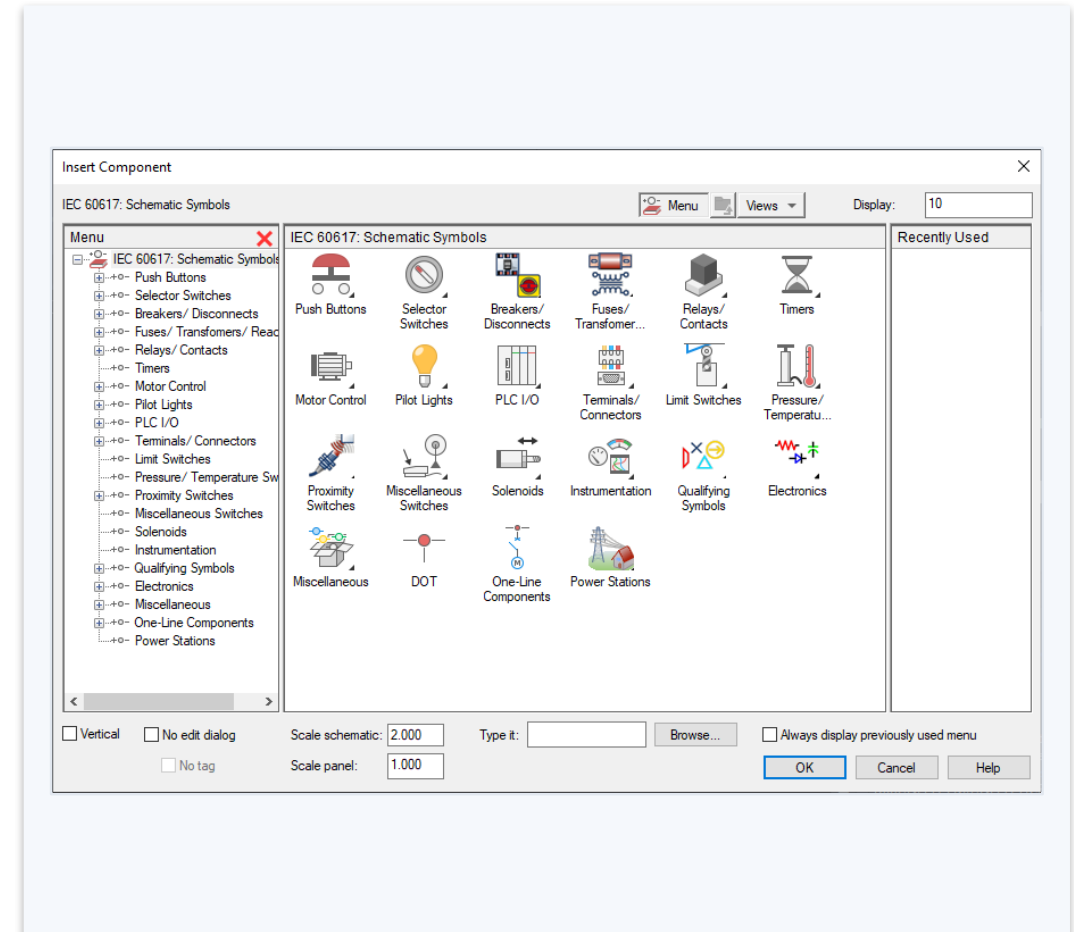
Insert a Component from the Catalog Database

- Know the part number already? Insert the component straight from the catalog database.
- Use the Catalog Browser palette; search values are saved for quick reuse.
- Catalog information lands on the symbol automatically; edit further in the Insert/Edit dialog that follows insertion.



Insert a Component from the Icon Menu

- The icon menu inserts by symbol type — a NO push button, a 3-position selector switch, and so on.
- Different icon menus exist per library set; choose the project's icon menu under Project Settings.
- After insertion, annotate with descriptions and catalog information.



Annotate the Symbol

- Each symbol carries a main catalog value plus optional multiple-catalog values.
- Edit a component to add, change or remove catalog assignments — Lookup opens the catalog search.
- Multiple Catalog attaches auxiliary parts (contact blocks, adapters) to the same device.
- Complete, accurate annotation is what makes reports and BOMs trustworthy downstream.

Insert / Edit Component

Component Tag
 CR406 ☐ fixed
 Use PLC Address
 Tags Used: Schematic Panel External List Options

Description
 Line 1: SAFETY
 Line 2: RELAY NO.1
 Line 3:
 List: Drawing Project Defaults Pick

Catalog Data
 Manufacturer: AB
 Catalog: 700-P200A1
 Assembly:
 Item: Count:
 Next>> 1 Previous
 Lookup Drawing Project
 Multiple Catalog
 Catalog Check

Cross-Reference
☐ Component override Setup
 Reference NO: NO/NC Setup
 Reference NC:

Installation code
 Drawing Project

Location code
 MCAB5
 Drawing Project
 Show/Edit Miscellaneous

Pins

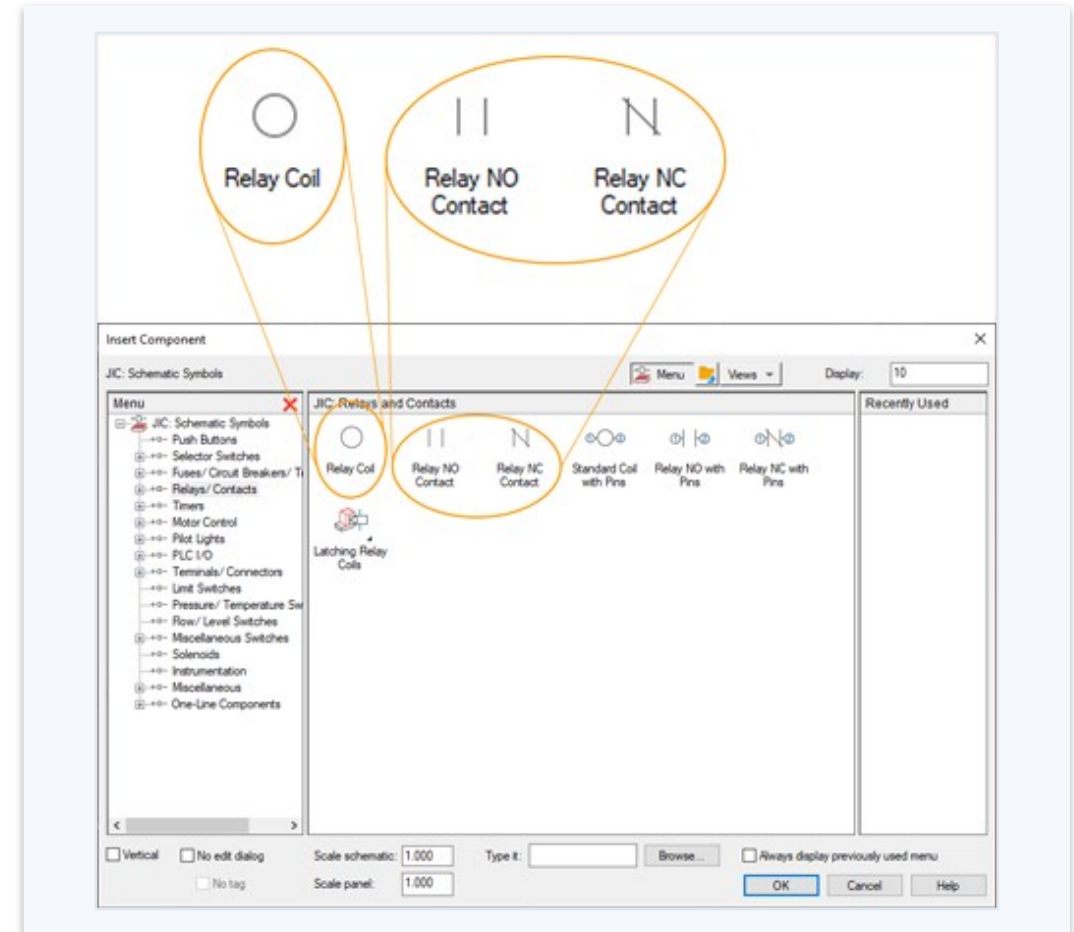
Pin	Terminal	Navigation
1	K1	< >
2	K2	< >
×		< >

Rating: Show All Ratings

OK-Repeat OK Cancel Help

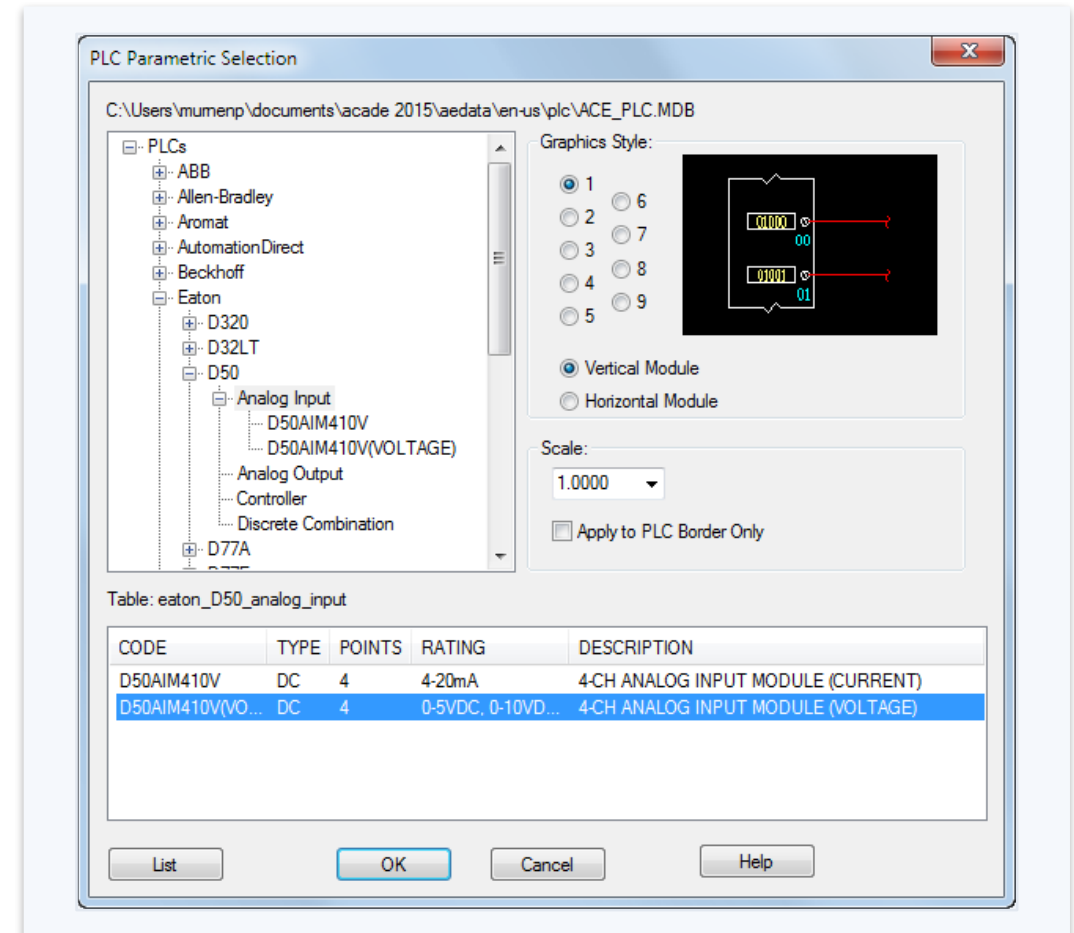
Parent and Child Components

- Some devices are drawn as several symbols — a relay is a coil plus its contact symbols.
- The coil is the parent; contacts are children that inherit its tag and data.
- When inserting a child, pick its parent from the active drawing or project list in the Insert/Edit Child dialog.
- Parent/child links power the cross-reference grid you will read in every review.



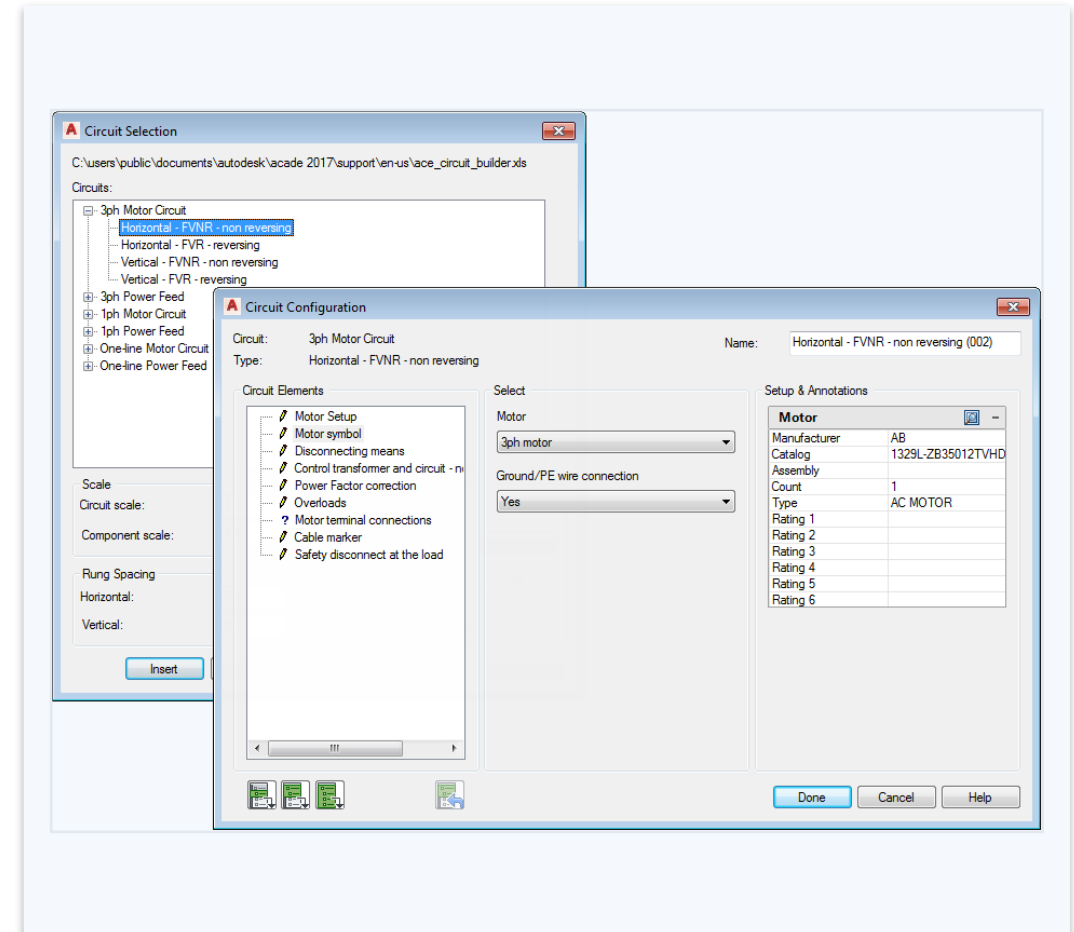
PLC I/O Modules

- AutoCAD Electrical generates hundreds of PLC I/O modules on demand, in multiple graphical styles.
- Modules adapt automatically to the underlying ladder rung spacing.
- Expand manufacturer → series → type, pick the module and style, then place it by insertion point.
- Parametric generation keeps PLC drawings consistent with the wiring standard in force.



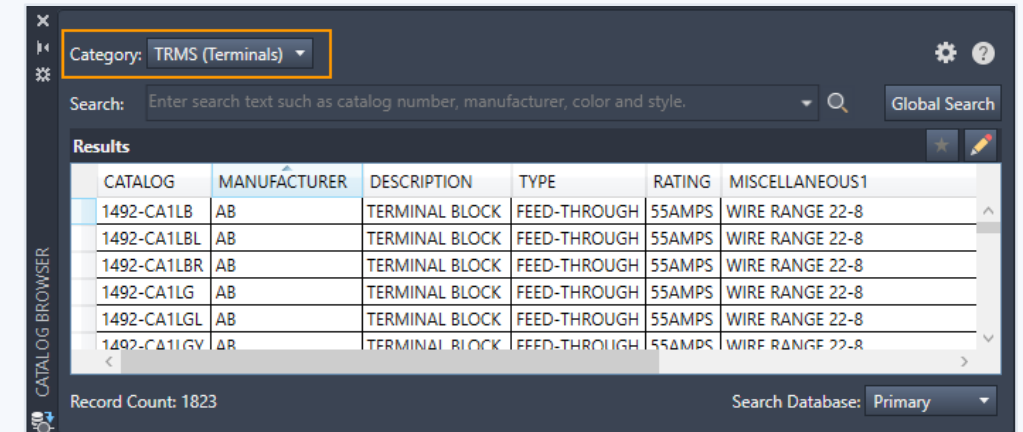
Circuits and Circuit Builder

- A circuit is any collection of components and wires; save frequently used circuits to the icon menu.
- A saved circuit re-inserts with tags updating automatically to stay unique.
- Circuit Builder assembles motor-control and power-feed circuits dynamically from your dialog selections.
- Building from proven circuits is how construction needs are met quickly and correctly.



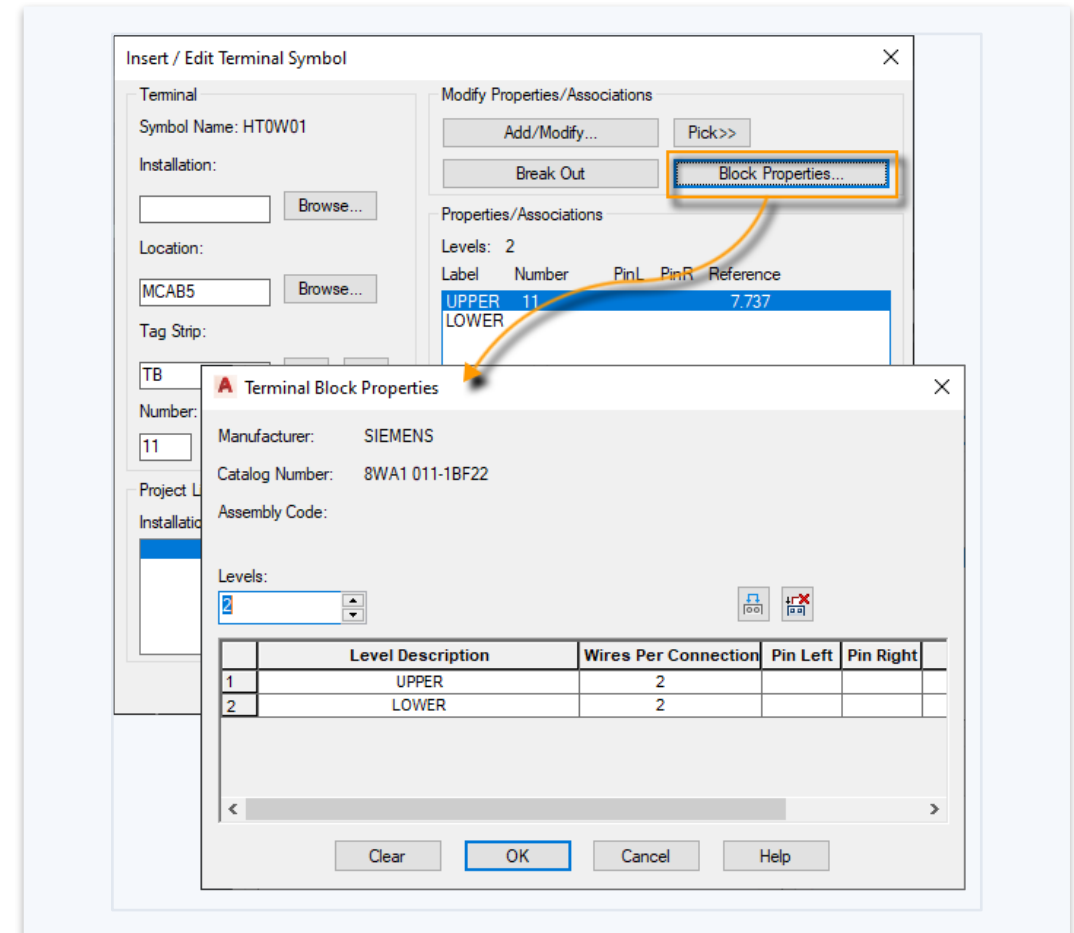
Schematic Terminals

- Insert terminals like other schematic symbols — from the Catalog Browser (TRMS category) or the icon menu.
- Four terminal behaviour types exist; the terminal block name controls the behaviour.
- Pick the behaviour type and terminal style carefully when inserting from the icon menu.
- The TERMINAL STRIP command inserts strip symbols using Installation, Location and Tagstrip values.



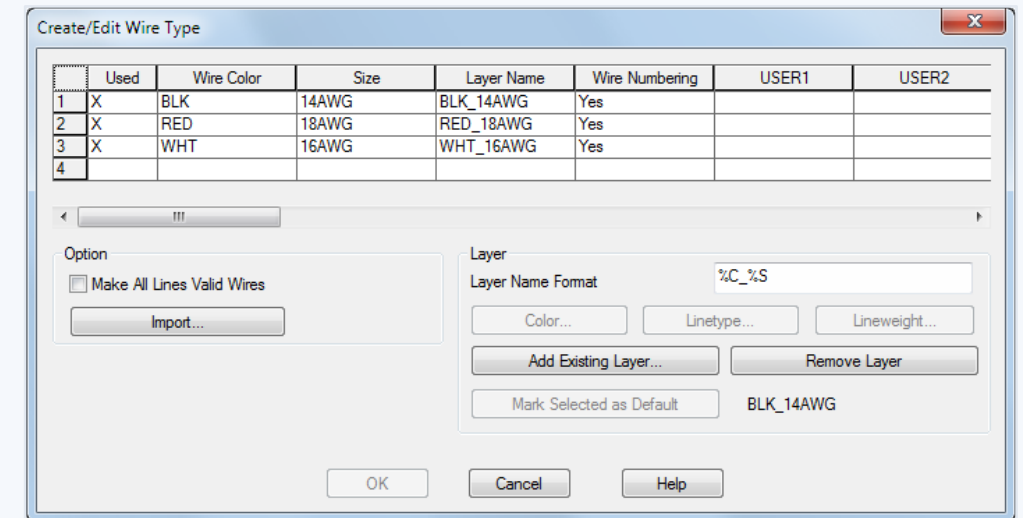
Terminal Block Properties

- A schematic terminal symbol represents a single-level terminal or ONE level of a multi-level terminal.
- The number of levels is a block property; each level carries its own wiring data.
- View and edit properties in the Insert/Edit Terminal Symbol dialog.
- Associate schematic terminals across levels — and manage whole strips in the Terminal Strip Editor.



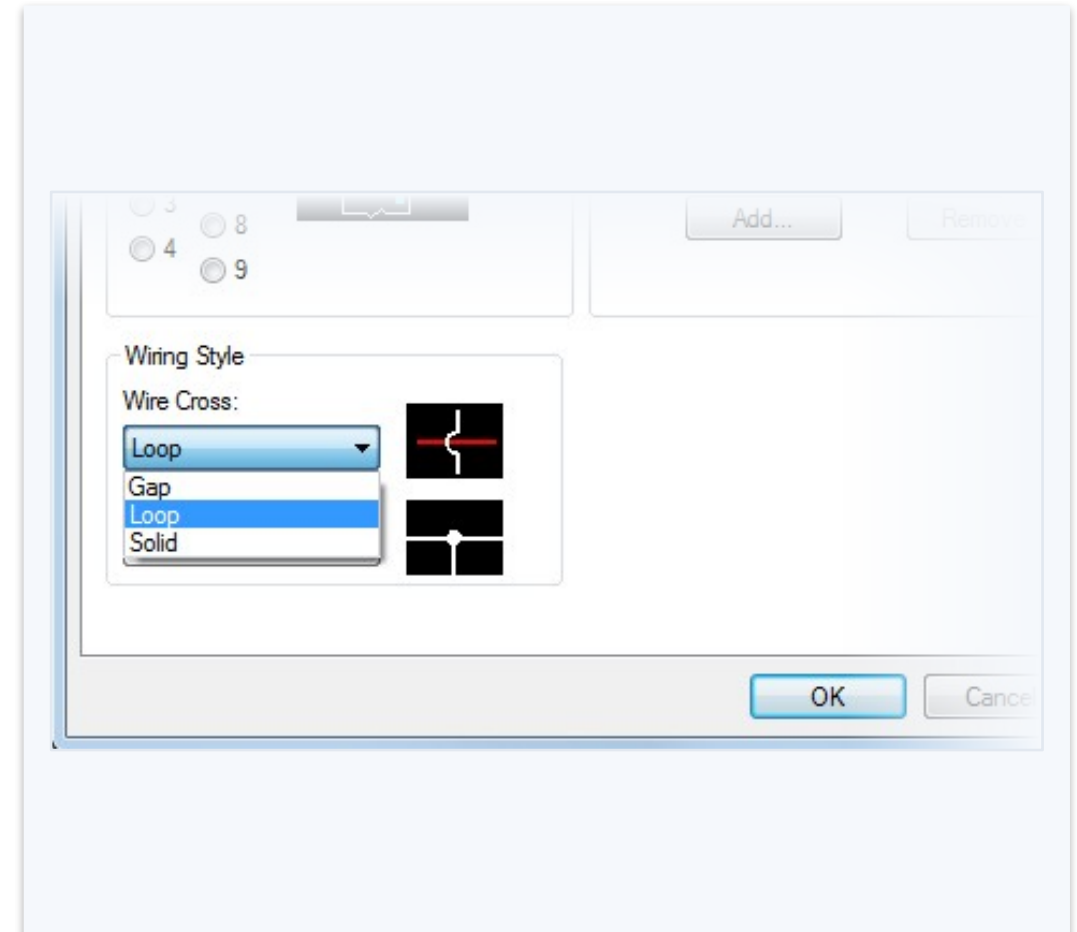
Wires and Wire Layers

- A wire is a line object on a layer defined as a wire layer; a wire network is connected wire segments.
- Manage wire layers in a grid: edit colour and size inline, right-click to rename.
- Create layers for wire families (24VDC, 230VAC, earth) so wiring intent is visible at a glance.



Insert Wires

- Inserted wires land on a wire layer automatically — joining an existing wire adopts that wire's layer.
- A tee connection receives a dot or angled tee per the Styles tab of Drawing Properties.
- Change/Convert Wire Type moves wires to another layer or converts plain lines into wires.

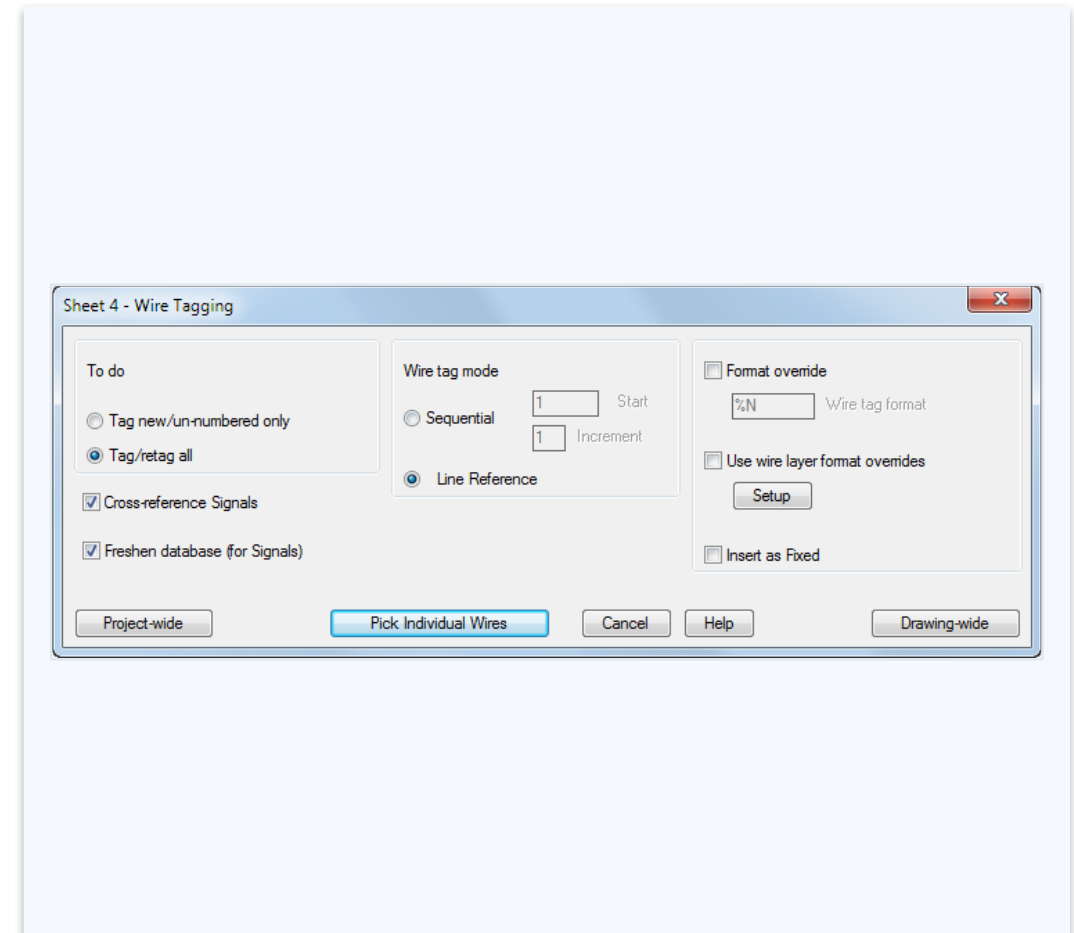


Wire Number Setup

- The Wire Numbers tab of Drawing Properties defines the wire-number format, type and placement.
- The format mixes text and replaceable parameters: %N (number), %S (sheet), %D (drawing), %X (suffix position), %P/%I/%L (IEC project / installation / location codes).
- Wire numbering is a specification — every sheet in the set must follow the same format.

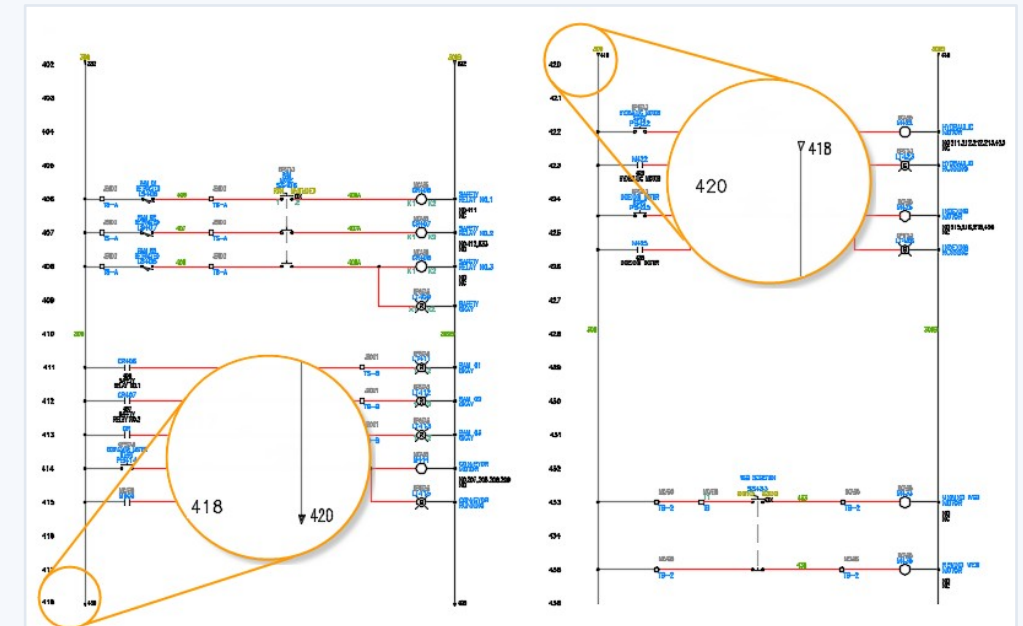
Insert Wire Numbers

- The Wire Numbers command inserts new numbers or updates existing ones drawing- or project-wide.
- Drawing-property settings are preselected; override per run if needed.
- Numbers marked Fixed are preserved by renumbering — edit them via Edit Wire Number.



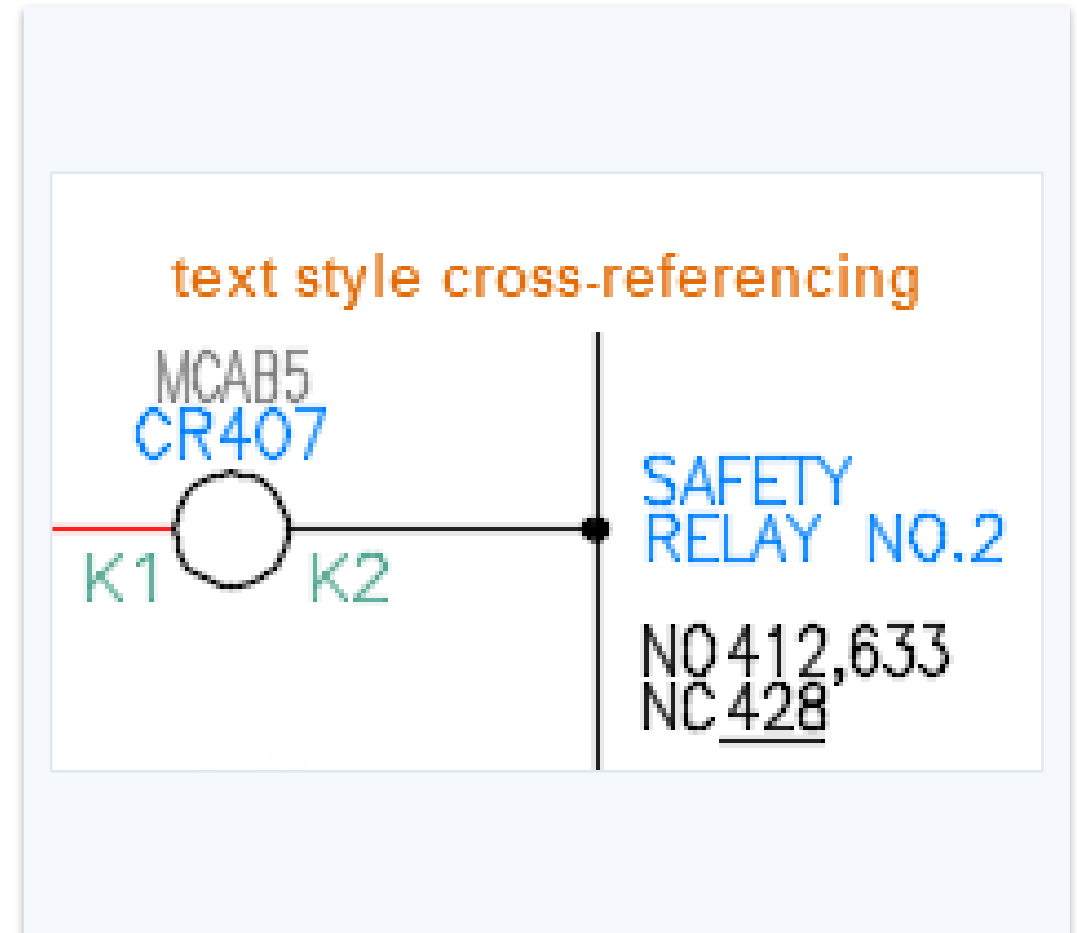
Signal Arrows

- A signal arrow breaks a wire network and continues it elsewhere — same drawing or another sheet.
- AutoCAD Electrical uses named source/destination pairs: give the source a code, then insert destinations that reference it.
- Arrows carry cross-reference annotation both ways, so a checker can trace the signal across the set.



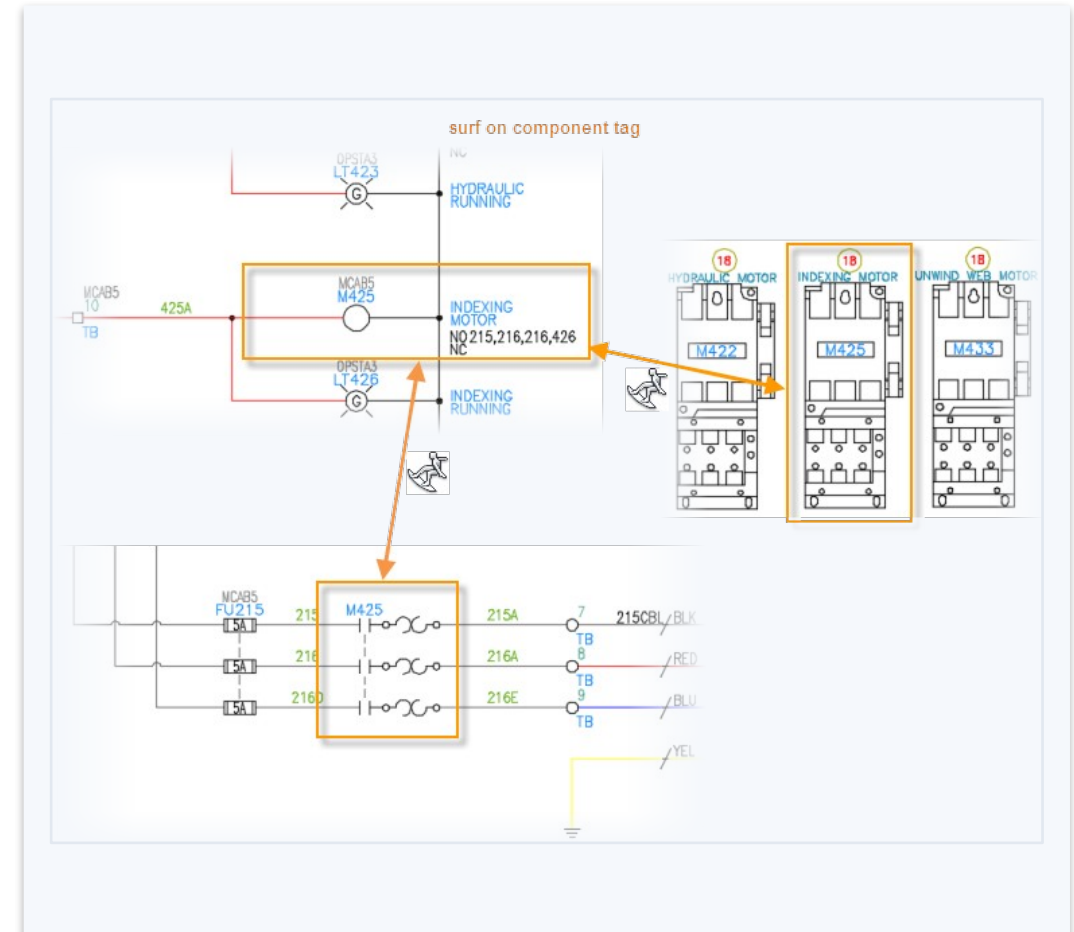
Cross-Referencing

- Cross-references annotate related components with each other's reference locations.
- Three annotation styles on a parent; child annotation is always text format.
- Signal arrows are cross-referenced too — source shows destinations, destination shows its source.
- Set the format and style on Drawing Properties → Cross-References; keep it current before any review or PDF publish.



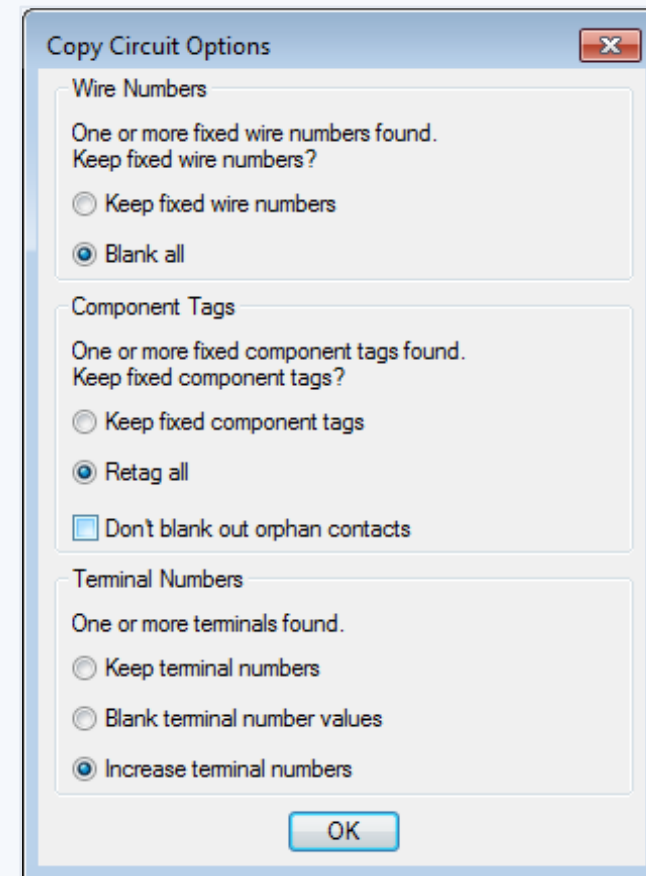
Surfing Between Components

- Surfer jumps between related components and follows signal arrows across drawings.
- Pick a parent and surf to each child, or follow a source arrow to its destinations.
- Surfing is the fastest way to audit that parents, children and signals all agree.



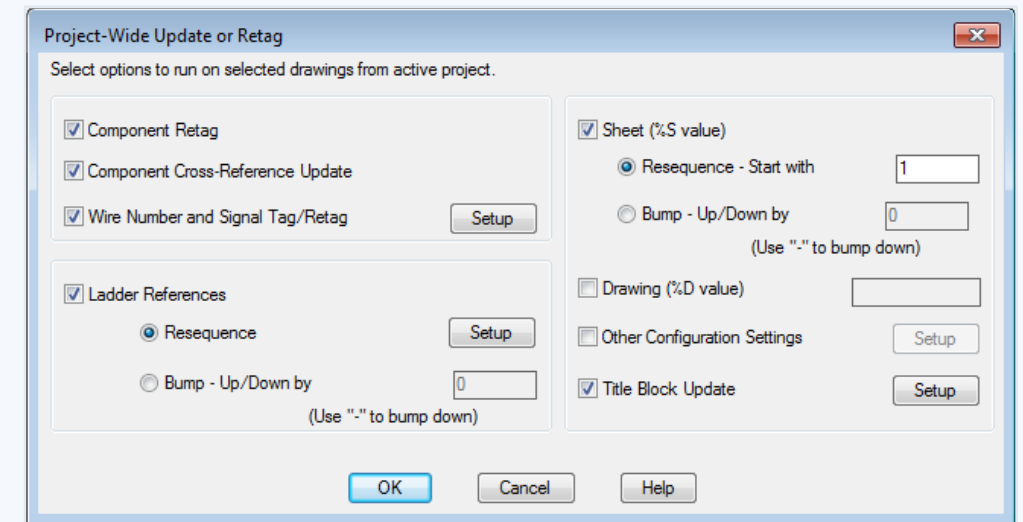
Modify Commands — Edit Without Breaking Intelligence

- Always modify with the Electrical editing commands so component intelligence survives.
- Align components; Move — tags update, wires heal and break as needed.
- Copy with multiple insertion points — tags stay unique; optionally update terminal numbers.
- Circuit Clipboard (Cut/Copy/Paste) moves components and wires between drawings.
- Quick edits: delete-and-heal, toggle NO/NC contact state, swap a block.



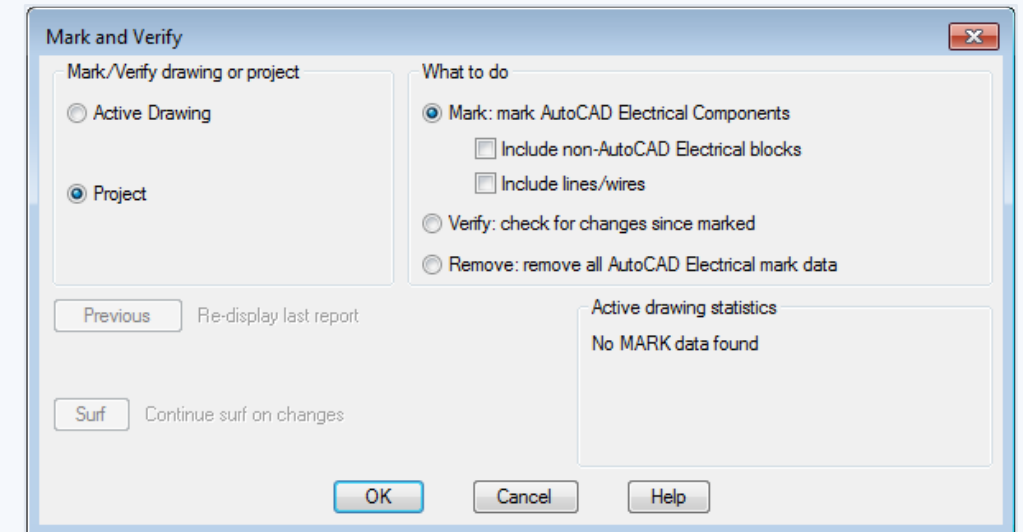
Project-Wide Updates

- Reusing drawings from a previous design? Update the reference system, retag components, reassign wire numbers and refresh cross-references in ONE pass.
- Project-Wide Update/Retag runs across every non-reference drawing in the project.
- This is what keeps a copied design consistent instead of quietly wrong.



Track Changes — Mark and Verify

- Mark/Verify tracks changes to your drawings — even changes made outside AutoCAD Electrical.
- Mark records the current state before edits begin.
- Verify afterwards reports every change — print it or surf straight to each changed object.
- A marked-and-verified set gives the reviewer an exact change list to check against the specification.



Hands-On Labs — Topic 2

Lab 3

- Build Circuits with Circuit Builder
- A schematic sheet containing a Circuit Builder motor circuit plus a saved, reusable circuit block.

Lab 4

- Insert Wires, Wire Numbers and Terminals
- A correctly wired and wire-numbered circuit with terminals, matching the drawing's wire-number specification.

Lab 5

- Cross-Reference and Surf a Drawing Set
- A cross-referenced two-sheet circuit you can navigate with Surfer, plus a Mark/Verify change report.

Build Circuits with Circuit Builder

LAB 3

Use Circuit Builder to insert a three-phase motor-control circuit built dynamically from your selections, then save a frequently used circuit and re-insert it from the icon menu.

You'll build

A schematic sheet containing a Circuit Builder motor circuit plus a saved, reusable circuit block.

Tools: Circuit Builder · Icon Menu · Schematic components · Saved circuits

Build Circuits with Circuit Builder

Test it

The motor circuit sits on the ladder with unique component tags (M1, PB1 ...), and your saved circuit re-inserts with automatically retagged components — construction needs are met without redrawing.

Insert Wires, Wire Numbers and Terminals

LAB 4

Insert wires between components, trim them, add automatic wire numbers in the drawing's specified format, and place schematic terminals — the accuracy checks every wiring diagram must pass.

You'll build

A correctly wired and wire-numbered circuit with terminals, matching the drawing's wire-number specification.

Tools: Wires · Wire Numbers · Trim Wire · Schematic Terminals · Power Plan DWG

Insert Wires, Wire Numbers and Terminals

Test it

Every wire carries a number in the drawing-property format, the fixed number survives a project-wide renumber, and the terminals join the wires cleanly — the sheet passes an accuracy review.

Cross-Reference and Surf a Drawing Set

LAB 5

Link a parent coil to its child contacts, break a wire across sheets with source/destination signal arrows, then audit the set with Surfer and Mark/Verify — how a checker reviews a real drawing set.

You'll build

A cross-referenced two-sheet circuit you can navigate with Surfer, plus a Mark/Verify change report.

Tools: Cross-referencing · Signal Arrows · Surfer · Mark/Verify · Reference DWG

Cross-Reference and Surf a Drawing Set

Test it

Surfer jumps coil → contact → signal arrow across sheets without manual searching, and the Verify report lists exactly the one component you moved — the set is auditable for accuracy.

Recap — Topic 2

- A4 — analyse drawings to determine construction needs; assemble standard motor-control circuits (Lab 3)
- A3 — review drawings for accuracy; wire the circuit and number it to specification (Lab 4)
- A3, A4 — review a multi-sheet set for accuracy with cross-references, Surfer and signal arrows (Lab 5)
- Key concepts: Types of technical drawings, Technical drawing standards, Schematic components, Terminals & PLC symbols, Wires, numbers & arrows, Review for accuracy.

SEE YOU TOMORROW · 0930

End of Day 1

Day 2 continues with drawing-set review, documentation, submission — and the final assessment.

Day 2

Digital attendance (AM) · Day 1 recap

TOPIC 03

Reviewing Technical Drawings for Strategic Relevance and Standard Compliance

Drafting documentation · panel layouts · reports & part lists · custom symbols · plot & publish for submission (K5, K6 · A5, A6)

Key Concepts — Topic 3

1

Fundamentals of documentation

Panel layouts, nameplates, item numbers and balloons turn a schematic into build documentation. (K5)

2

Reports & part lists

BOM, wire from/to and component reports extract project data for checking and procurement.

3

Custom symbols

Symbol Builder creates house-standard symbols so drawings match the organisation's conventions. (K6)

4

Strategic relevance

Reuse projects, track changes and align drawing sets with the organisation's design strategies. (A5)

5

Standard compliance

Title blocks, xrefs and plot standards keep sheets conformant to submission requirements. (A6)

6

Plot & publish

Output the project to a printer or a hyperlinked multi-sheet PDF for stakeholder submission.

Fundamentals of Drafting Design Documentation

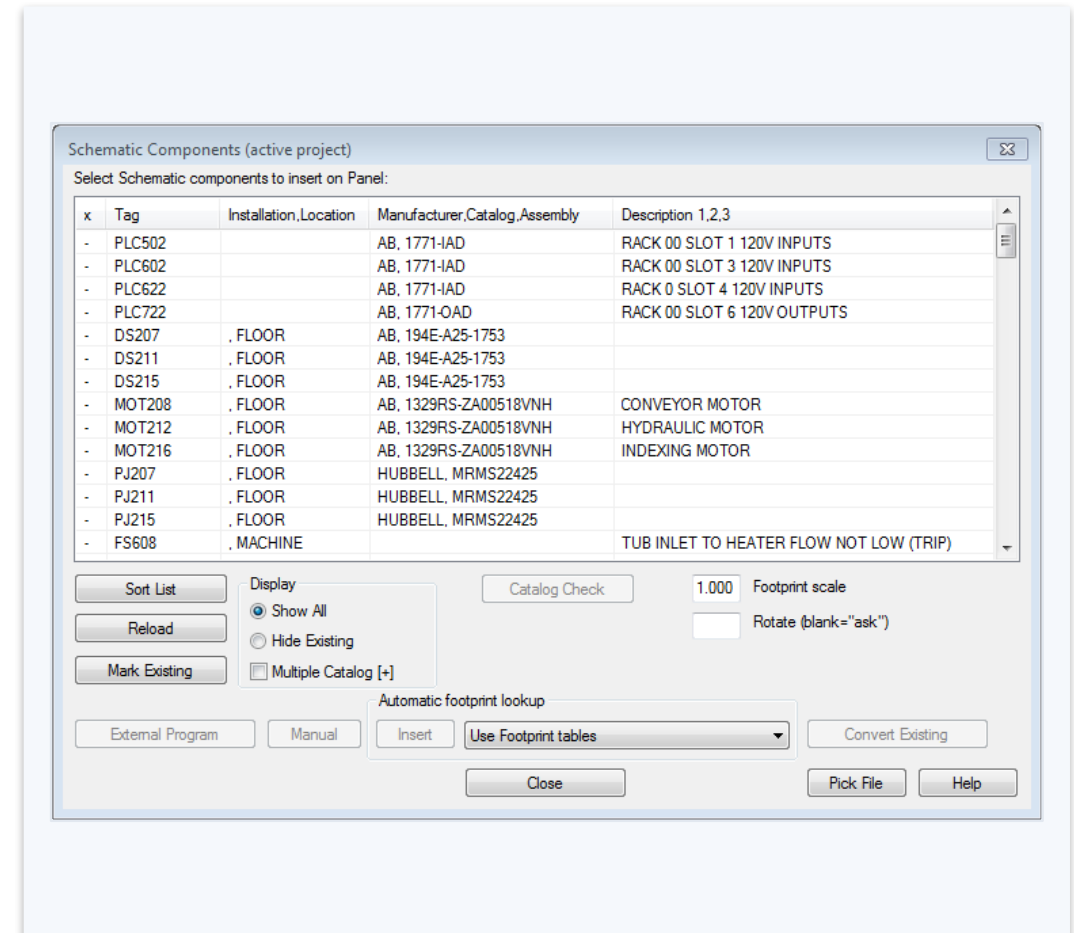
- A schematic alone cannot be built: documentation adds the physical and procurement views.
- Panel layouts show where each device mounts; nameplates label them; item numbers and balloons key parts to the BOM.
- Reports extract part lists, wire lists and terminal plans from the same project data — no retyping.
- Documentation must align with the organisation's strategies (house symbols, formats) and submission standards. (A5, A6)

Panel Layouts

- Insert panel footprint symbols from the schematic component list or the panel icon menu.
- The PANEL ribbon tab carries the layout tools; drawing properties hold panel-specific settings.
- Add components and connect with the insert tools; nameplates, item numbers and balloons complete the layout.

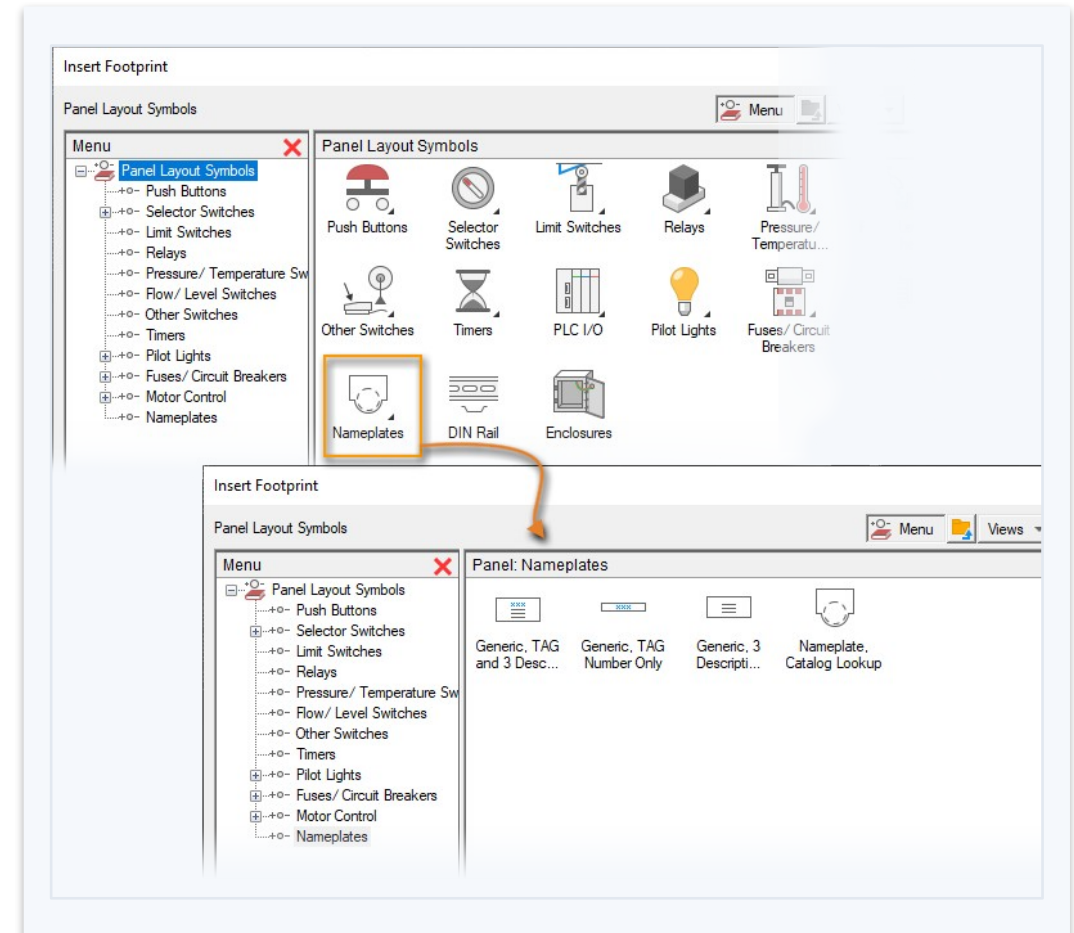
Insert Footprints from the Schematic List

- Draw schematics first, then build the panel by selecting from the schematic component list.
- The catalog assignment finds the correct footprint; browse or draw one on the fly if missing.
- The dialog shows which components are already placed; all values copy from schematic to footprint.
- Change a value on either side later and the other side updates — schematic and panel stay consistent.



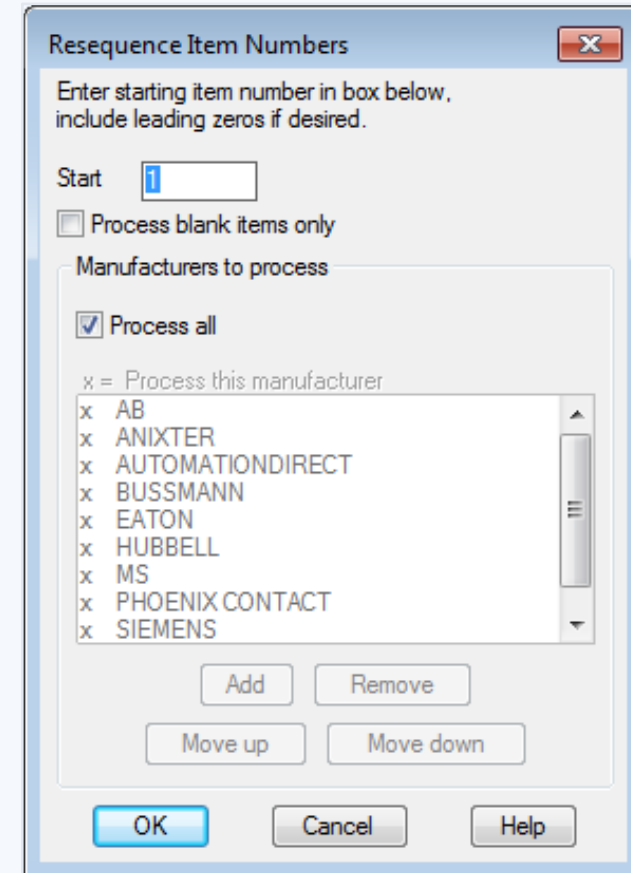
Panel Icon Menu and Nameplates

- Components not on the schematic (wire duct, rails) insert from the Panel Icon Menu.
- Insert nameplates from the icon menu and associate each with a footprint.
- Tag, installation, location and descriptions copy from the footprint — override if required.



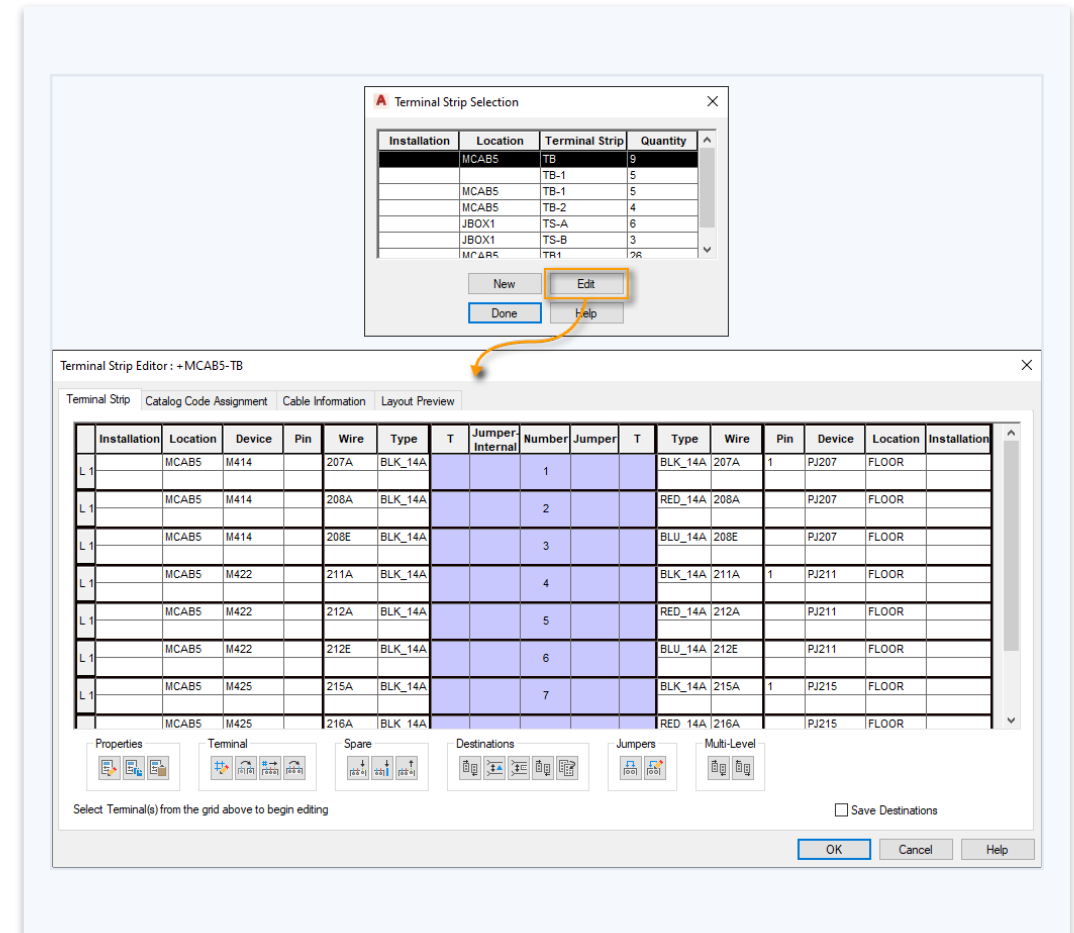
Item Numbers and Balloons

- Item numbers key parts to the BOM; components sharing a catalog value share an item number.
- Panel components are numbered first, then schematic components.
- Set the item-numbering project properties before resequencing.
- Insert balloons to make item numbers visible; multi-catalog components prompt for which value to balloon.



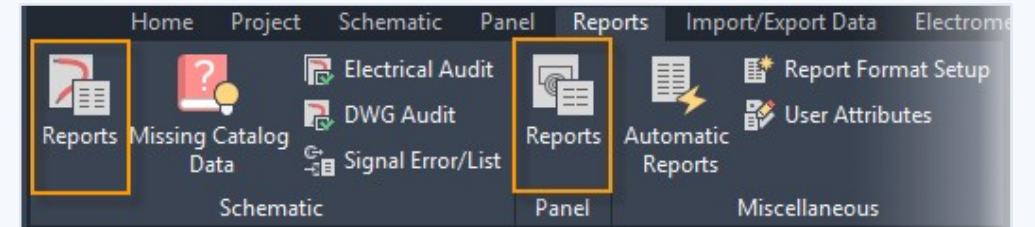
Terminal Strip Editor

- Manage an entire terminal strip at once: catalog values, connected components, wires and numbers.
- Resequence terminal numbers, reorder terminals, add spares and accessories, insert jumpers.
- The strip list merges panel terminals and schematic terminals into one editable view.
- Insert the strip graphically — or generate tabular terminal strips with the Table Generator.



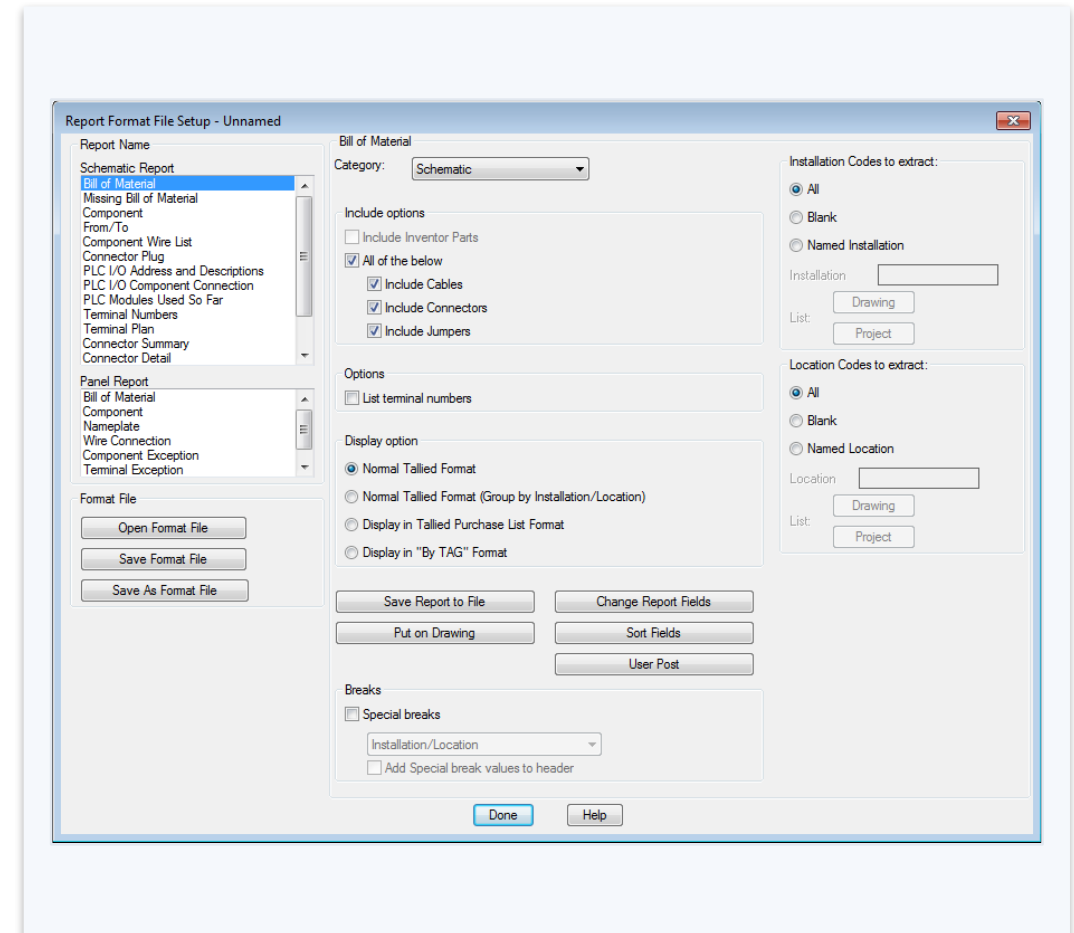
Reports

- Generate project reports: Bill of Material, component lists, wire from/to, terminal plans and more.
- The REPORT command builds custom reports from drawing data; select the report, options and drawings to process.
- Output to a file for procurement, or insert on the drawing as a table for the construction set.



Report Fields and Format Files

- After running a report, include/exclude fields, reorder them and rename labels.
- Save the settings as a .SET format file — unlimited format files per report type.
- Format files ARE the organisation's reporting standard: everyone reproduces the same layout in one click.
- Automatic Reports runs a whole group of formatted reports in a single pass.



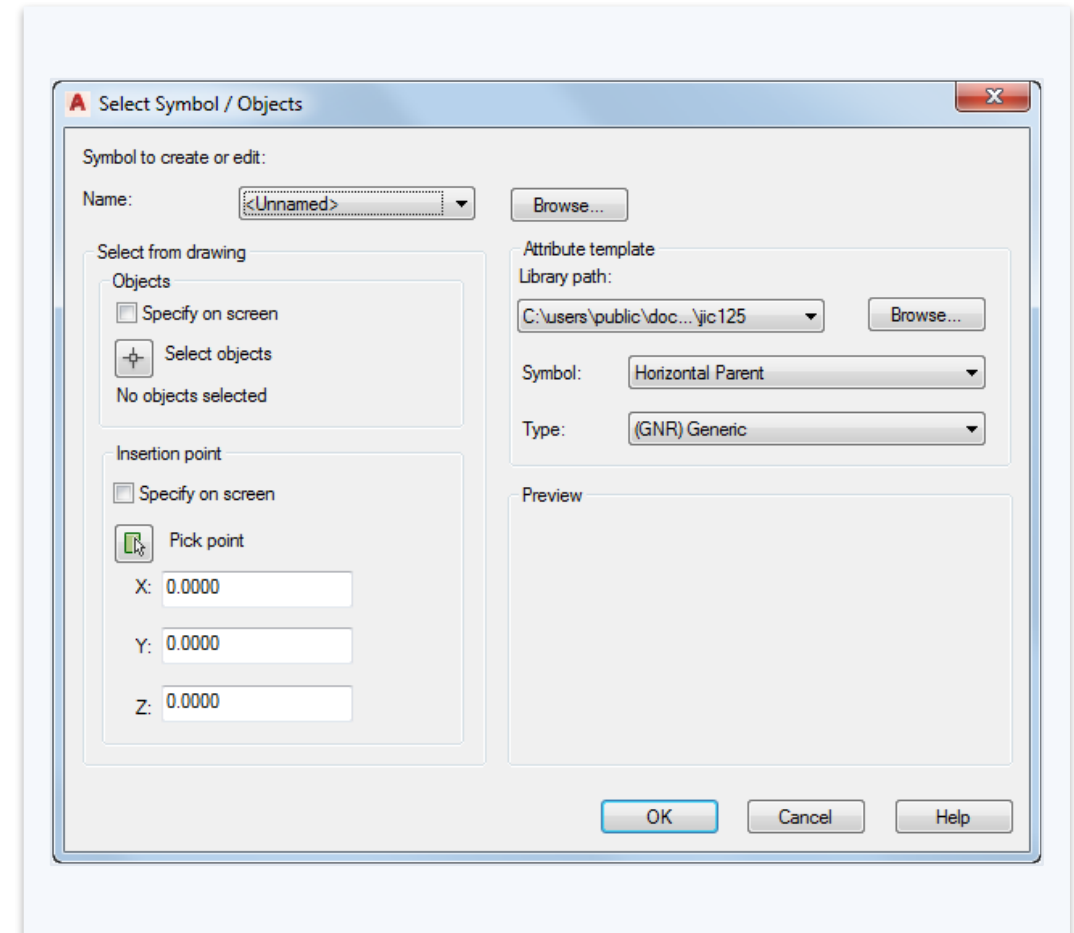
Output a Report — File or Drawing Table

- Save to File exports the report (CSV, XLS, MDB) for procurement and checking.
- Put on Drawing inserts the report as an AutoCAD table — style it with TABLESTYLE.
- The drawing table and the exported file come from the same data, so they cannot disagree.

Report for Project VDDEMO					
TAGS	QTY	SUB	CATALOG	MFG	DESCRIPTION
CR406 CR407 CR408	3		700-P200A1	AB	P TYPE ELECTRICALLY HELD RELAY, 4 POLE, AC COIL, 115-120VAC 60Hz / 110VAC 50Hz COIL, RATING 10A, OPEN TYPE P 120VAC 2 NO
PB414	1	*1	800EP-E2	AB	PUSH BUTTON - MOMENTARY, IP66, NEMA 4/4X/13 22.5mm EXTENDED, IEC STYLE BLACK 2 NO PLASTIC OPERATOR w 2 ACROSS MTG
		*2	800E-2X10	AB	800E CONTACT BLOCK 22.5mm IEC STYLE 1 NO
		*1	800E-A2L	AB	800E MOUNTING LATCH - 2 ACROSS 22.5mm IEC STYLE 2 ACROSS MOUNTING LATCH 2 ACROSS MOUNTING LATCH
LT415 LT423 LT426	3		800H-PR16G	AB	GREEN PILOT LIGHT - STANDARD, ROUND DILTIGHT 30.5mm 120VAC XFMR PLASTIC LENS CORROSION RESISTANT
LT409 LT411 LT412 LT413	4		800H-PR16R	AB	RED PILOT LIGHT - STANDARD, ROUND DILTIGHT 30.5mm 120VAC XFMR PLASTIC LENS CORROSION RESISTANT
SS406 SS433	2		800MR-HH2BB	AB	SELECTOR SW - 2 POS MAINT, NEMA 13 22.5mm BLACK KNOB 2 NO 2 NC

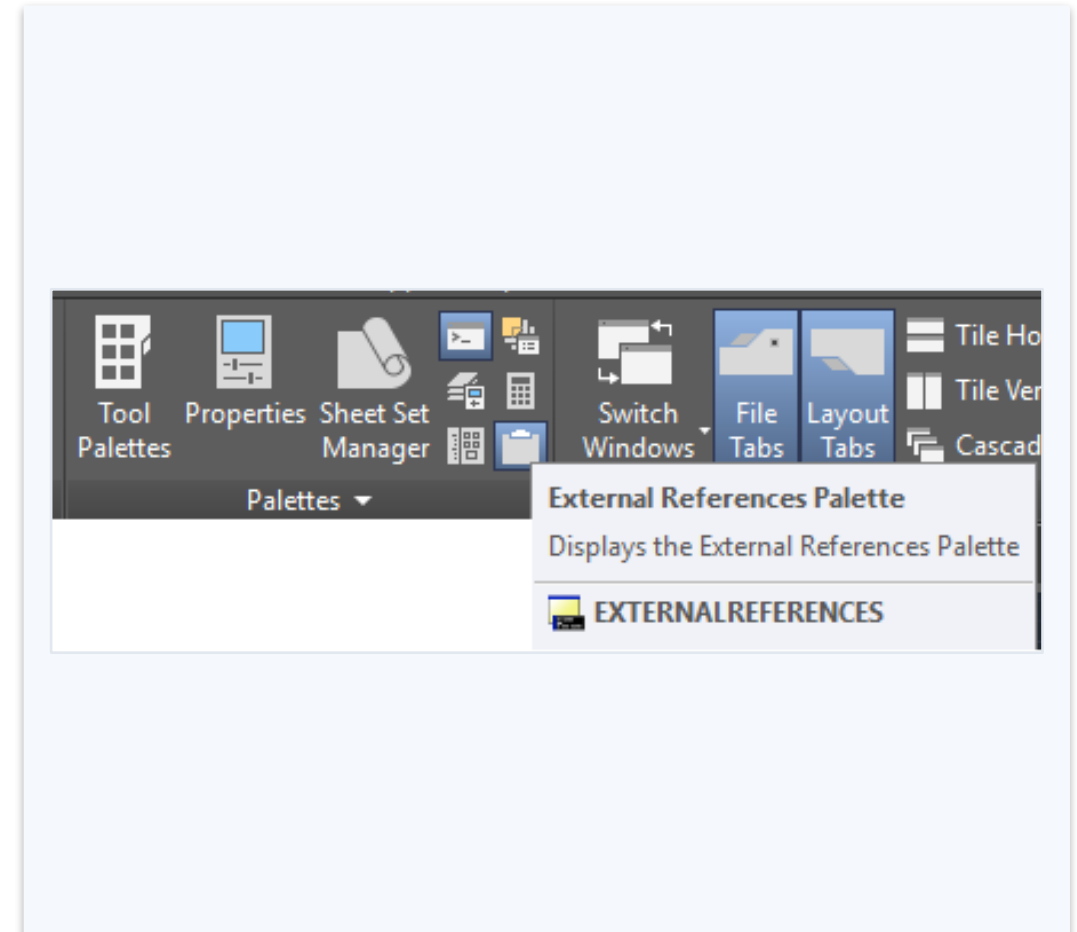
Custom Symbols with Symbol Builder

- Symbol Builder creates custom symbols fully compatible with AutoCAD Electrical.
- Start from scratch or from a similar existing symbol; set symbol type, insertion point and orientation.
- For schematic symbols create both the parent and child versions.
- Draw the graphics, then save following the AutoCAD Electrical symbol naming convention — that is what makes the symbol intelligent.



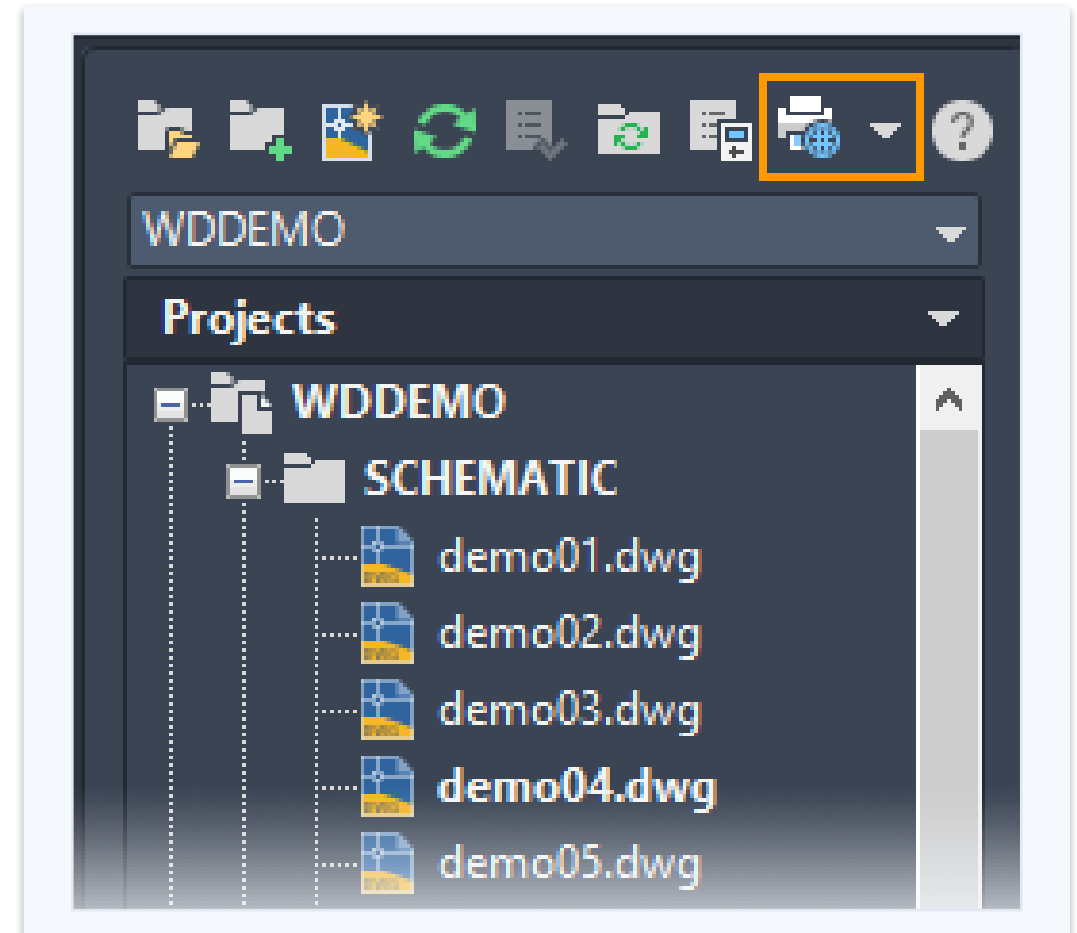
External References (Xrefs)

- An external reference attaches another file into the current drawing without copying it.
- The External References palette organises DWG, DWF, DGN, PDF and image references.
- Attach via the palette (View tab → Palettes) or the XREF command; pick the file and placement.
- Xrefs keep shared content (title blocks, outlines) in ONE file — edit once, update everywhere.



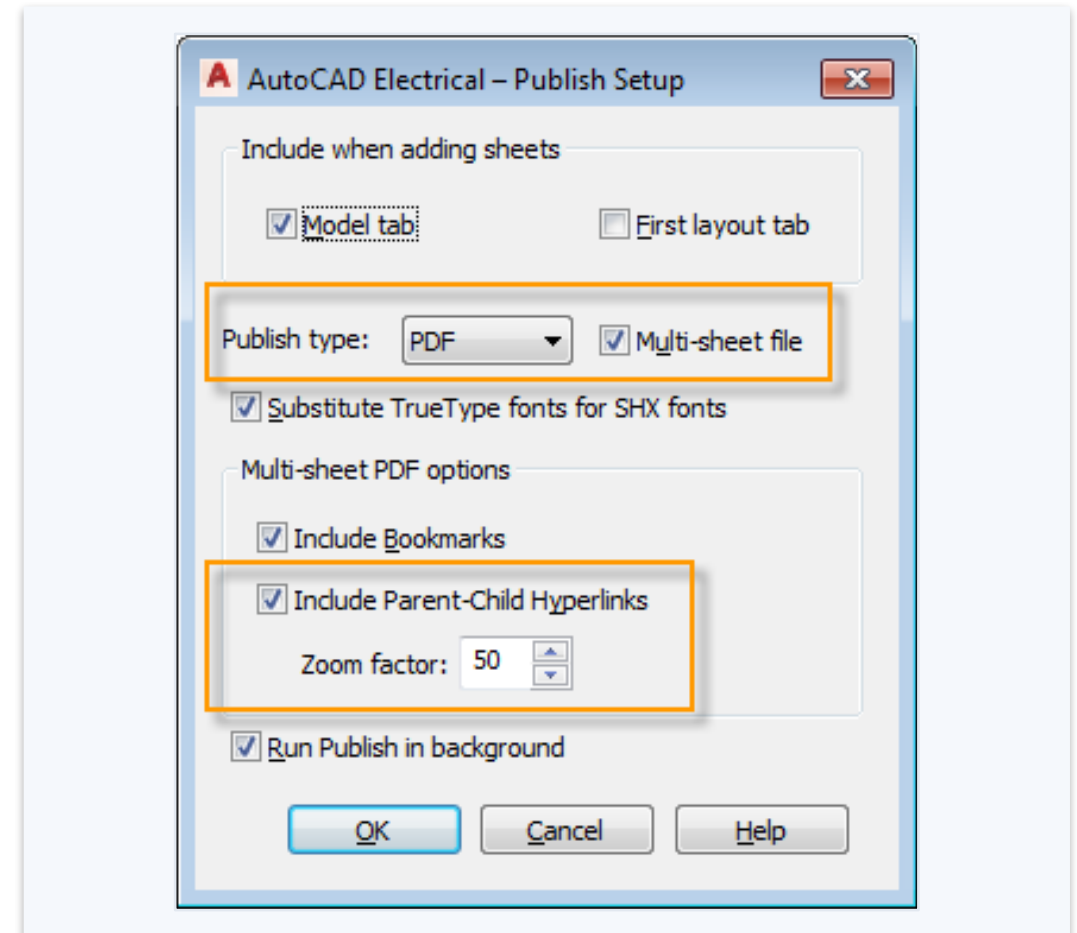
Plot a Project

- Print an entire project, a subset of sheets, or a single drawing.
- Use Publish/Plot on Project Manager → Plot Project, then pick the drawings to output.
- Select the layout to plot per drawing; use a named page setup for consistent output.
- Map a plot date/time attribute in the title block — it stamps automatically at plot time.



Publish to a Hyperlinked PDF

- Publish the project to a multi-sheet PDF that preserves relationships as hyperlinks.
- Tick Multi-sheet file and Include Hyperlinks; the zoom factor sets how close a clicked link zooms.
- Each object carries one hyperlink — reach a child through the parent's cross-reference text.
- Update cross-referencing BEFORE publishing; then reviewers can navigate the set without AutoCAD.



Hands-On Labs — Topic 3

Lab 6

- Create a Panel Layout with Footprints
- A panel layout drawing with footprints placed from the schematic list, a nameplate, item numbers and balloons.

Lab 7

- Generate Reports and Part Lists
- A BOM table placed on the drawing plus a saved .SET format file and a CSV export of the report.

Lab 8

- Plot and Publish the Project to PDF
- An A3 PDF plot of one sheet and a hyperlinked multi-sheet PDF of the whole project.

Create a Panel Layout with Footprints

LAB 6

Generate a panel layout from the schematic component list, insert footprints and a nameplate, and assign item numbers with balloons — drafting design documentation the workshop can build from.

You'll build

A panel layout drawing with footprints placed from the schematic list, a nameplate, item numbers and balloons.

Tools: Panel tab · Schematic List · Footprints · Nameplates · Item balloons · Blocks DWGs

Create a Panel Layout with Footprints

Test it

Each footprint shows the same tag as its schematic parent, the nameplate carries the footprint's description, and balloons display consistent item numbers — the panel documents the design faithfully.

Generate Reports and Part Lists

LAB 7

Run Bill of Material and Wire From/To reports across the project, tailor the fields, save a format file for repeat use, and insert the report as a table on the drawing — the part lists a checker reviews.

You'll build

A BOM table placed on the drawing plus a saved .SET format file and a CSV export of the report.

Tools: Reports · Bill of Material · Format files (.SET) · Tables · Attributes DWG

Generate Reports and Part Lists

Test it

The BOM table on the drawing matches the CSV export line for line, and re-running the report through your .SET format file reproduces the same layout — the part list is standard-compliant and repeatable.

Plot and Publish the Project to PDF

LAB 8

Attach an external reference, then plot a drawing to an A3 PDF and publish the whole project to a single multi-sheet PDF with hyperlinks — the final conformance step before submission.

You'll build

An A3 PDF plot of one sheet and a hyperlinked multi-sheet PDF of the whole project.

Tools: External References · Plot · Publish · Multi-sheet PDF · Layout tabs DWG

Plot and Publish the Project to PDF

Test it

The A3 PDF matches the layout exactly, and in the multi-sheet PDF clicking a cross-reference hyperlink jumps to the referenced component — the set is ready for standards-conformant submission.

Recap — Topic 3

- A5 — turn the schematic into build documentation aligned with the organisation's design strategy (Lab 6)
- A6, K5 — extract BOM and wire lists, format them to house standards and place them on the drawing (Lab 7)
- A6 — output the drawing set to submission standards (A3 sheets, hyperlinked multi-sheet PDF) (Lab 8)
- Key concepts: Fundamentals of documentation, Reports & part lists, Custom symbols, Strategic relevance, Standard compliance, Plot & publish.



WRAP-UP

Course Summary & Assessment

What You Achieved

1

Set up to standard

Configured drawings from templates with limits, units, grid and snap — production metrics in force. (A1)

2

Projects & drawings

Built .wdp projects, managed drawing order, properties and reference-only sheets. (A1, A2)

3

Schematic drawing

Inserted components, circuits, terminals and PLC modules to IEC/IEEE/NFPA standards. (K1, K3)

4

Wired & numbered

Wired circuits with layered wires, spec-format wire numbers and signal arrows. (K4, A3)

5

Reviewed drawing sets

Audited accuracy with cross-references, Surfer and Mark/Verify. (A3, A4)

6

Documented & submitted

Produced panel layouts, BOM reports, custom symbols and hyperlinked PDF submissions. (K5, K6, A5, A6)

Recommended Courses

1

WSQ - Technical Drawing with AutoCAD

2

WSQ - Technical Drawing with AutoCAD
Mechanical

3

WSQ - Application of BIM using Revit

4

WSQ - Architecture Drawing with Revit

5

WSQ - Product Design with Fusion 360

6

WSQ - AutoCAD Civil 3D for Infrastructure
Design

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg

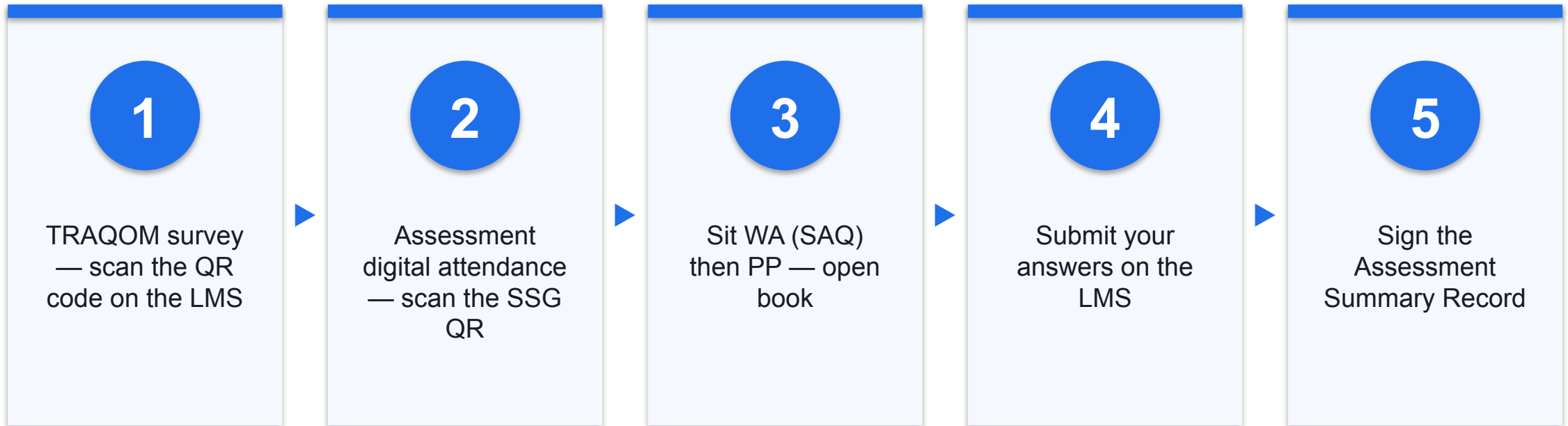
Cert & TRAQOM Survey (Mandatory)

- Complete the course feedback and the mandatory SSG TRAQOM survey.
- Your e-certificate is issued via the LMS/TMS portal: <https://lms-tms.tertiaryinfotech.com>
- The trainer will display the TRAQOM QR code — please complete it before the assessment.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. (K1–K6)
- Practical Performance (PP) — hands-on AutoCAD Electrical tasks, 1 hour, open book. (A1–A6)
- Format: Open Book — slides, Learner Guide and approved materials only.
- Submit your completed scripts to the assessor and sign the Assessment Summary Record.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

HAPPY DRAFTING

Thank You!

You are now ready to produce, analyse and review technical drawings with AutoCAD Electrical.